03 - Minimax

Pattern Recognition and Machine Learning, Lab 3

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Test next week

Test: 16 October

- At the **beginning of the lab**
- Topics: **Bayesian** and **non-Bayesian** tasks
- Theory and worked problems
- Also things from the **assignments**, e.g. what a Python function computes
- Bring a **calculator**
- No phone, laptop or notes

Bonus tasks

- Handed in **in person** after the lab
- You **explain** the task and convince us that the solution is correct

Cannot come to the test? Let me know in advance.

Recap

Recap: Bayesian decision making

What do we need to know to find the Bayesian strategy?

$$R(q) = \sum_{x \in X} \sum_{k \in K} p_{XK}(x, k)\, W(k, q(x)), \qquad q^* = \arg \min_{q\colon X \to D} R(q)$$

We need

- $p(x \mid k)$: likelihoods
- $p(k)$: **priors**
- $W(k, d)$: the **loss**

With the 0/1 loss

$$\begin{aligned} q^*(x) &= \arg\max_k p(k \mid x) \\ &= \arg\max_k p(x \mid k)\, p(k) \end{aligned}$$

The prior is inside the posterior.

Recap: the reject option

When is it better to answer "I do not know"?

Decisions $D = K \cup \{\text{not known}\}$ and the loss

$$W(k, d) = \begin{cases} 0 & \text{if } d = k \\ 1 & \text{if } d \neq k \text{ and } d \neq \text{not known} \\ \varepsilon & \text{if } d = \text{not known} \end{cases}$$

The Bayesian strategy becomes

$$q^*(x) = \begin{cases} \arg\max_k p(k \mid x) & \text{if } 1 - \max_k p(k \mid x) < \varepsilon \\ \text{not known} & \text{otherwise} \end{cases}$$

When Bayes is not enough

Priors are unknown

A plate reader used **anywhere** in the country: every crossroad has different $p(k)$.

The loss is hard to set

What is the price of an overlooked tumour compared with a false alarm?

A guarantee is required

"At most 1 % of dangerous states may be overlooked."

? What can we still do when we know only the likelihoods $p(x \mid k)$?

Non-Bayesian tasks

Error on class k: $\epsilon(k) = \sum_{x:\, q(x) \neq k} p(x \mid k)$

Neyman-Pearson

$$\min_{q} \ \epsilon(2)$$

subject to $\epsilon(1) \leq \bar{\epsilon}$

1 dangerous, 2 normal: few false alarms, overlooked danger bounded

Wald

$$\min_{q:\, X \to D} \ \max_{k \in \{1,2\}} \ \kappa(k)$$

subject to

$\epsilon(1) \leq \epsilon, \ \epsilon(2) \leq \epsilon$

decisions $D = K \cup \{?\}$, the answer "?" is no error

undecided: $\kappa(k) = \sum_{x:\, q(x) = ?} p(x \mid k)$

Minimax

$$\min_{q} \ \max_{k \in K} \ \epsilon(k)$$

the worst class is as good as possible

today

For two classes, all three are solved by **thresholding the likelihood ratio** $r(x) = \frac{p(x|1)}{p(x|2)}$

Discrete minimax

The minimax task

$$q^* = \arg \min_{q\colon X \to K} \max_{k \in K} \epsilon(k), \qquad \epsilon(k) = \sum_{x\colon q(x) \neq k} p(x \mid k)$$

? What is $\epsilon(k)$ in words? What do we need, and what is missing compared with Bayes?

For two classes: decide by the likelihood ratio

$$r(x) = \frac{p(x \mid 1)}{p(x \mid 2)}, \qquad q(x) = \begin{cases} 1 & \text{if } r(x) > \lambda \\ 2 & \text{if } r(x) \leq \lambda \end{cases}$$

One number gives the whole strategy: the threshold λ.

Lecture: the inverse ratio $p(x \mid 2)/p(x \mid 1)$ with the threshold μ. Assignment: this form.

Discrete example: female or male?

Observation x = (height, weight), $K = \{1, 2\} = \{F, M\}$ (female, male)

$p(x \mid F)$

	u-light	light	avg	heavy
short	.197	.145	.094	.017
normal	.077	.299	.145	.017
tall	.001	.008	.000	.000

$p(x \mid M)$

	u-light	light	avg	heavy
short	.011	.005	.011	.011
normal	.005	.071	.408	.038
tall	.002	.014	.255	.169

? 1. What is $r(x) = \frac{p(x|M)}{p(x|F)}$ for each cell?

? 2. How do we find the minimax strategy q^*?

? 3. What are its errors $\epsilon(F)$ and $\epsilon(M)$?

Derivation: the likelihood ratio and its order

$r(x) = p(x \mid M) \,/\, p(x \mid F)$

	u-light	light	avg	heavy
short	0.056	0.034	0.117	0.647
normal	0.065	0.237	2.814	2.235
tall	2.000	1.750	∞	∞

Index i after sorting by r (from 0)

	u-light	light	avg	heavy
short	1	0	3	5
normal	2	4	9	8
tall	7	6	10	11

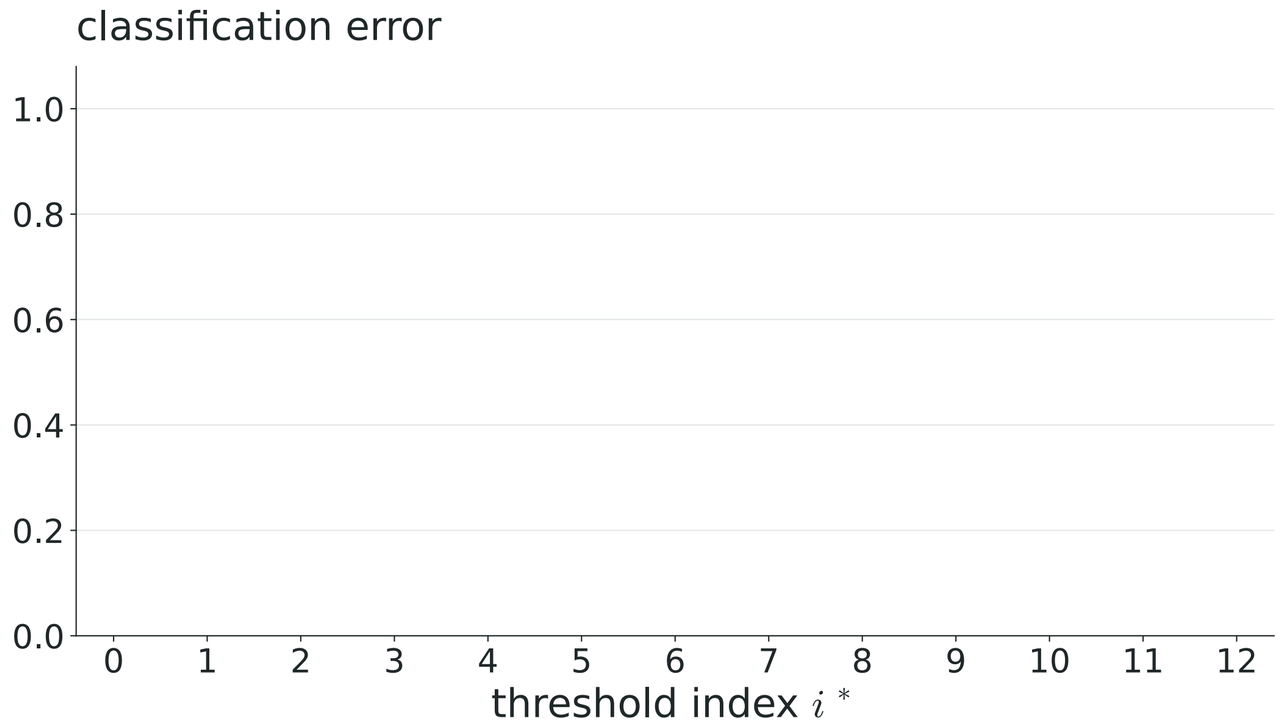
Every threshold on r is one index $i^* \in \{0, 1, \ldots, 12\}$:

$$q(x_i) = \begin{cases} M & \text{if } i \geq i^* \\ F & \text{if } i < i^* \end{cases}$$

$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F), \qquad \epsilon(M) = \sum_{i < i^*} p(x_i \mid M)$$

Derivation: errors for every threshold

threshold i^*	0	1	2	3	4	5	6	7	8	9	10	11	12
M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$													
$\epsilon(M)$													
max													



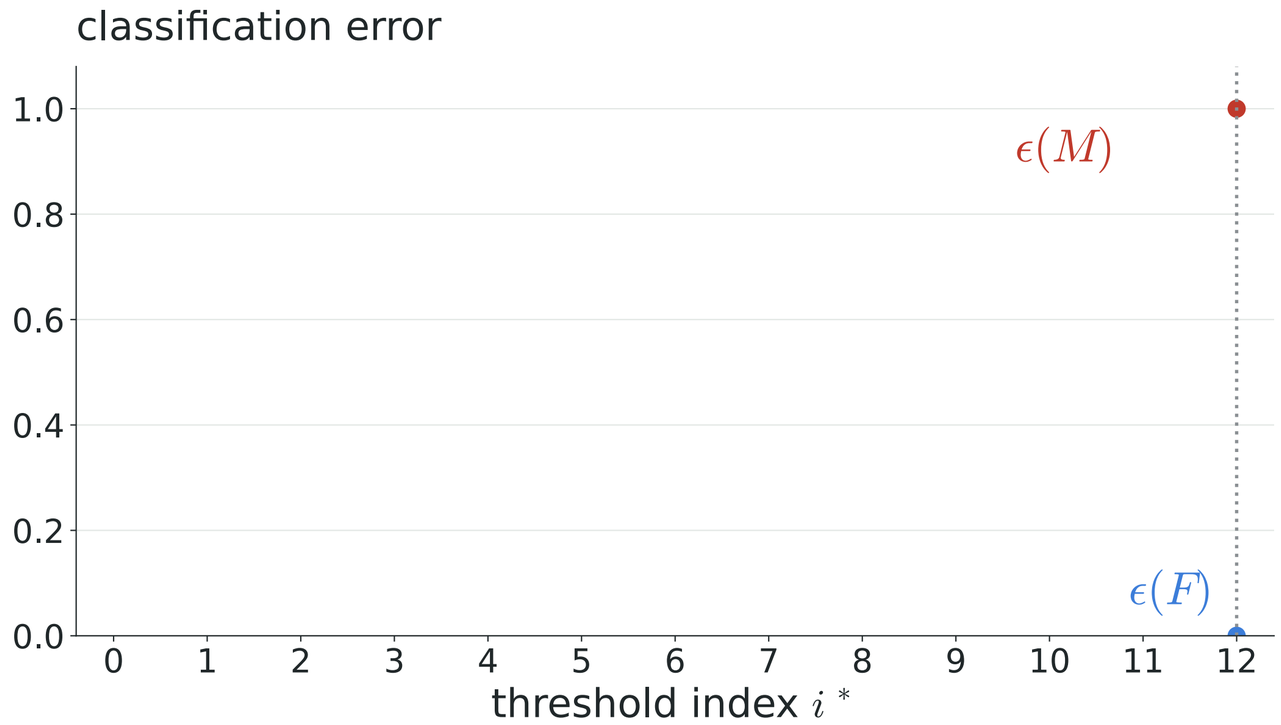
Sorted by r, decide M for $i \geq i^*$:

$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F)$$

$$\epsilon(M) = \sum_{i < i^*} p(x_i \mid M)$$

Derivation: errors for every threshold

threshold i^*	0	1	2	3	4	5	6	7	8	9	10	11	12
M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$													0
$\epsilon(M)$													1
max													1



Sorted by r, decide M for $i \geq i^*$:

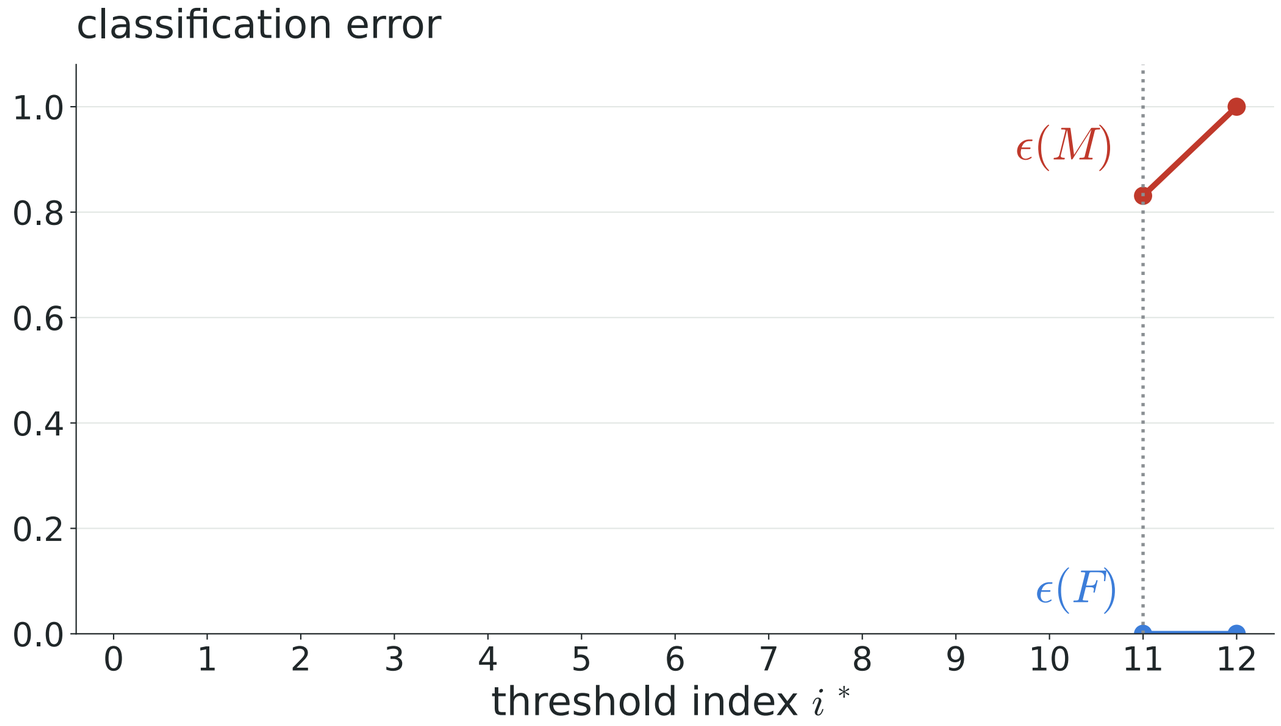
$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F)$$

$$\epsilon(M) = \sum_{i < i^*} p(x_i \mid M)$$

$i^* = 12$: nothing is decided M, i.e. we **always decide F**.

Derivation: errors for every threshold

threshold i^*	0	1	2	3	4	5	6	7	8	9	10	11	12
M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$												0	0
$\epsilon(M)$												.831	1
max												.831	1



Sorted by r, decide M for $i \geq i^*$:

$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F)$$

$$\epsilon(M) = \sum_{i < i^*} p(x_i \mid M)$$

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M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$											0	0	0
$\epsilon(M)$											.576	.831	1
max											.576	.831	1



Sorted by r, decide M for $i \geq i^*$:

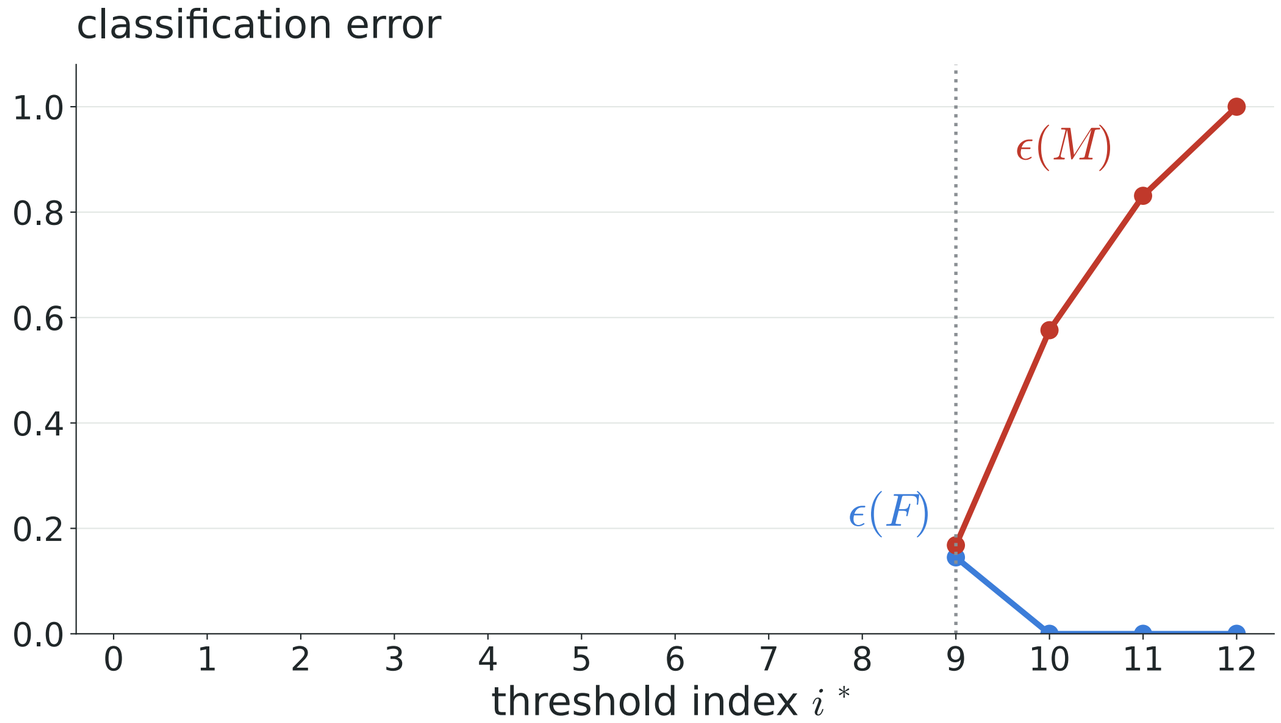
$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F)$$

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$\epsilon(F)$										.145	0	0	0
$\epsilon(M)$										.168	.576	.831	1
max										.168	.576	.831	1



Sorted by r, decide M for $i \geq i^*$:

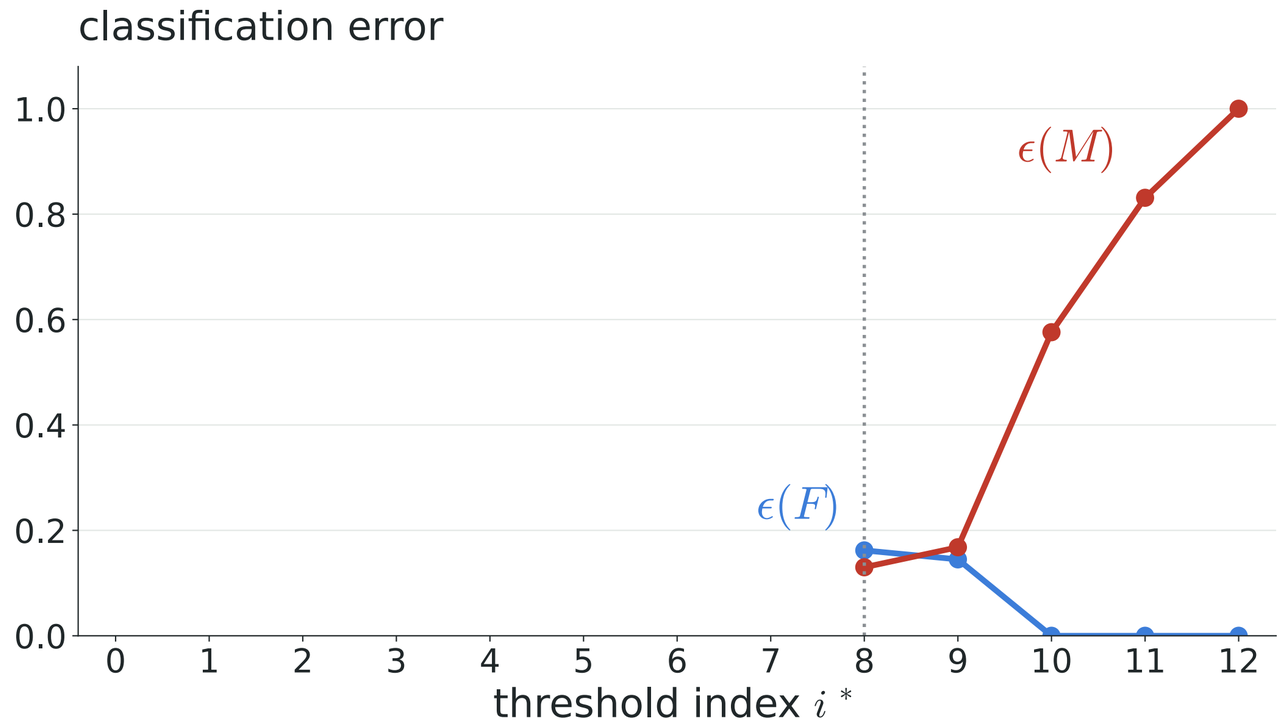
$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F)$$

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Derivation: errors for every threshold

threshold i^*	0	1	2	3	4	5	6	7	8	9	10	11	12
M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$									.162	.145	0	0	0
$\epsilon(M)$									.130	.168	.576	.831	1
max									.162	.168	.576	.831	1



Sorted by r, decide M for $i \geq i^*$:

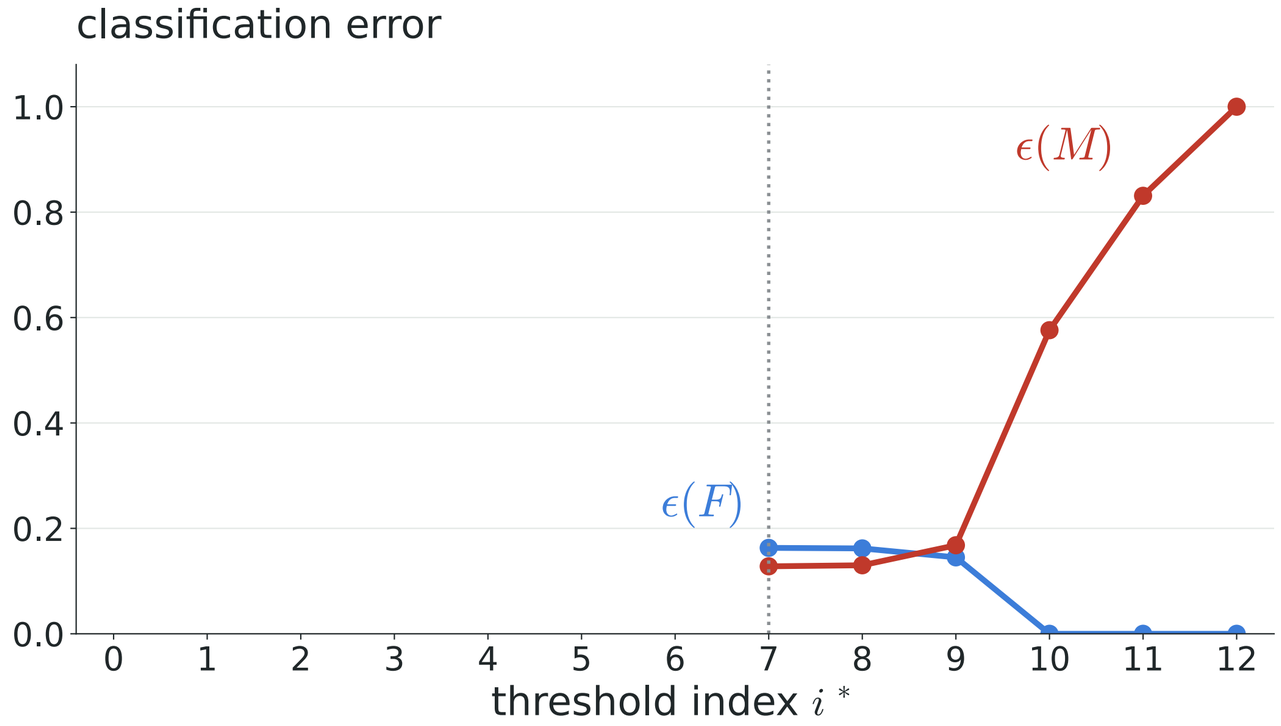
$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F)$$

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$i^* = 12$: nothing is decided M, i.e. we **always decide F**.

Derivation: errors for every threshold

threshold i^*	0	1	2	3	4	5	6	7	8	9	10	11	12
M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$								.163	.162	.145	0	0	0
$\epsilon(M)$								.128	.130	.168	.576	.831	1
max								.163	.162	.168	.576	.831	1



Sorted by r, decide M for $i \geq i^*$:

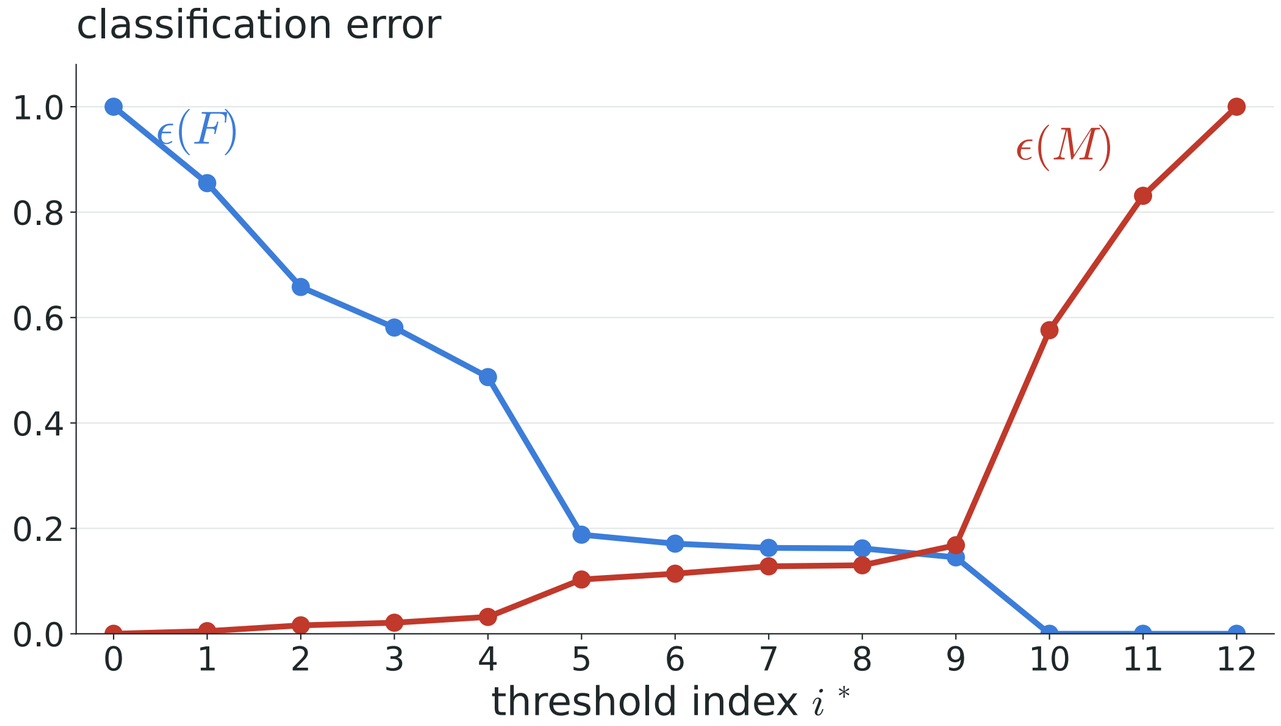
$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F)$$

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$i^* = 12$: nothing is decided M, i.e. we **always decide F**.

Derivation: errors for every threshold

threshold i^*	0	1	2	3	4	5	6	7	8	9	10	11	12
M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$	1	.855	.658	.581	.487	.188	.171	.163	.162	.145	0	0	0
$\epsilon(M)$	0	.005	.016	.021	.032	.103	.114	.128	.130	.168	.576	.831	1
max	1	.855	.658	.581	.487	.188	.171	.163	.162	.168	.576	.831	1



Sorted by r, decide M for $i \geq i^*$:

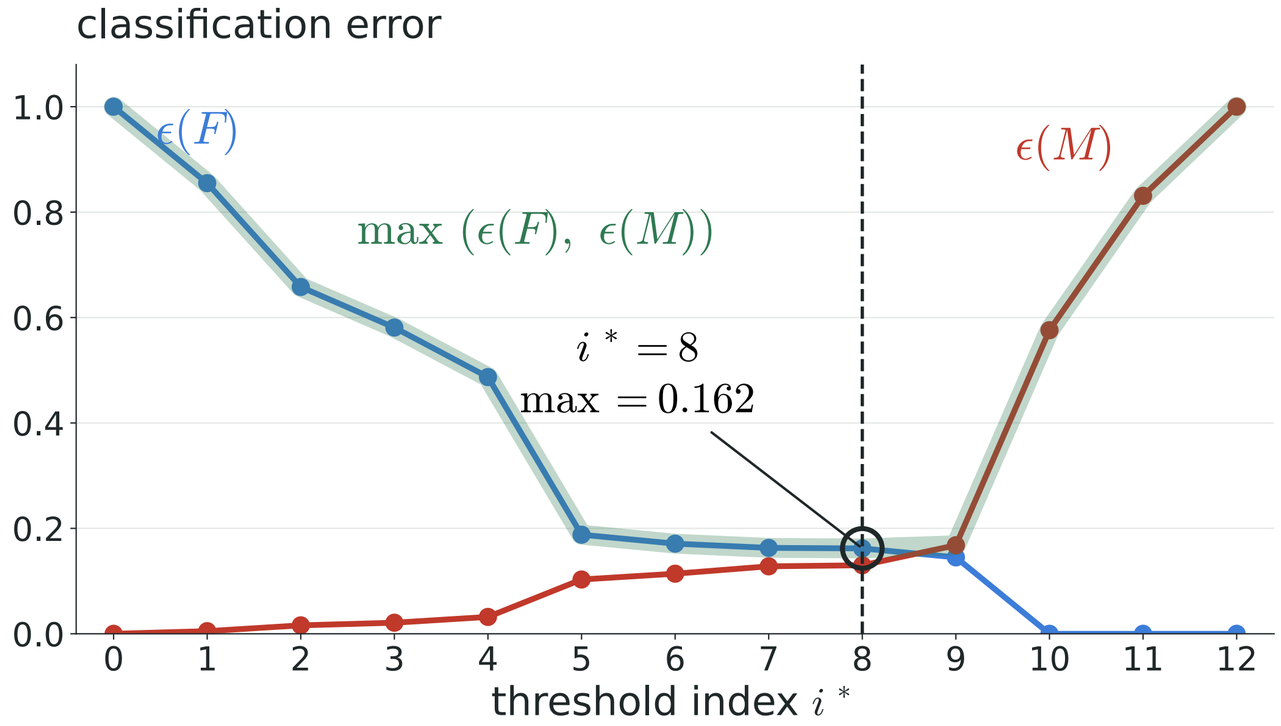
$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F)$$

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Derivation: errors for every threshold

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M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$	1	.855	.658	.581	.487	.188	.171	.163	.162	.145	0	0	0
$\epsilon(M)$	0	.005	.016	.021	.032	.103	.114	.128	.130	.168	.576	.831	1
max	1	.855	.658	.581	.487	.188	.171	.163	**.162**	.168	.576	.831	1



Sorted by r, decide M for $i \geq i^*$:

$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F)$$

$$\epsilon(M) = \sum_{i < i^*} p(x_i \mid M)$$

$i^* = 12$: nothing is decided M, i.e. we **always decide F**.

Smallest maximum: $\mathbf{i^* = 8}$

Derivation: the discrete minimax strategy

Strategy $q^*(x)$ for $i^* = 8$

	u-light	light	avg	heavy
short	F	F	F	F
normal	F	F	M	M
tall	F	F	M	M

$$q^*(x) = \begin{cases} M & \text{if } r(x) \geq 2.235 \\ F & \text{otherwise} \end{cases}$$

$$\epsilon(F) = 0.162, \quad \epsilon(M) = 0.130, \quad \max = \mathbf{0.162}$$

> ! For discrete x the errors usually **cannot be made equal**: take the smallest maximum.

Assignment: ratios 0, ∞, nan and ties are common. Thresholding the **index** handles them (nan: prefer the first class).

Continuous minimax

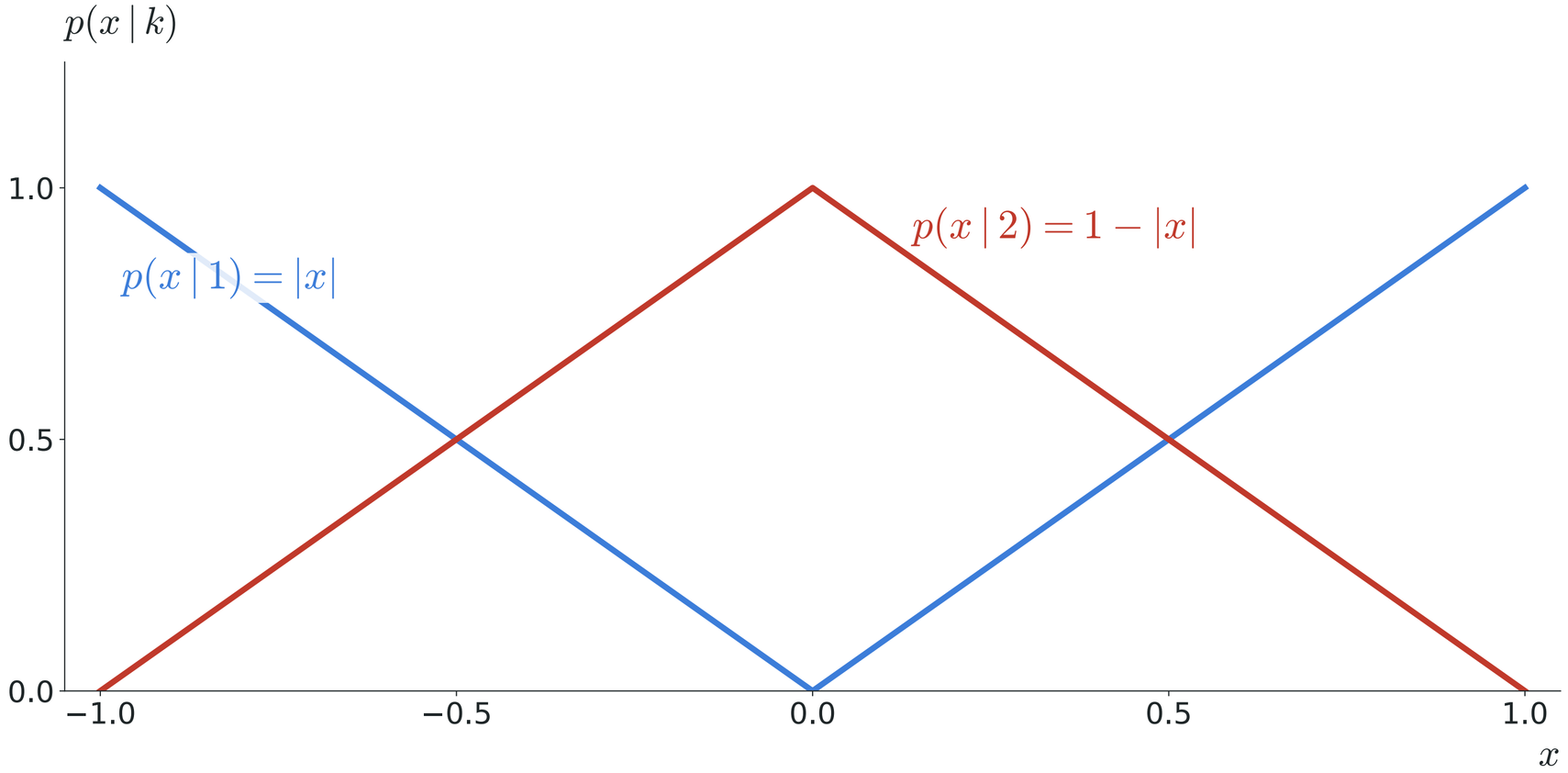
Minimax for continuous x

$$q^* = \arg \min_{q\colon X \to K} \max_{k \in K} \epsilon(k), \qquad \epsilon(k) = \int_{x\colon q(x) \neq k} p(x \mid k)\, dx$$

❓ We move the threshold λ. What happens to $\epsilon(1)$, $\epsilon(2)$ and their maximum?

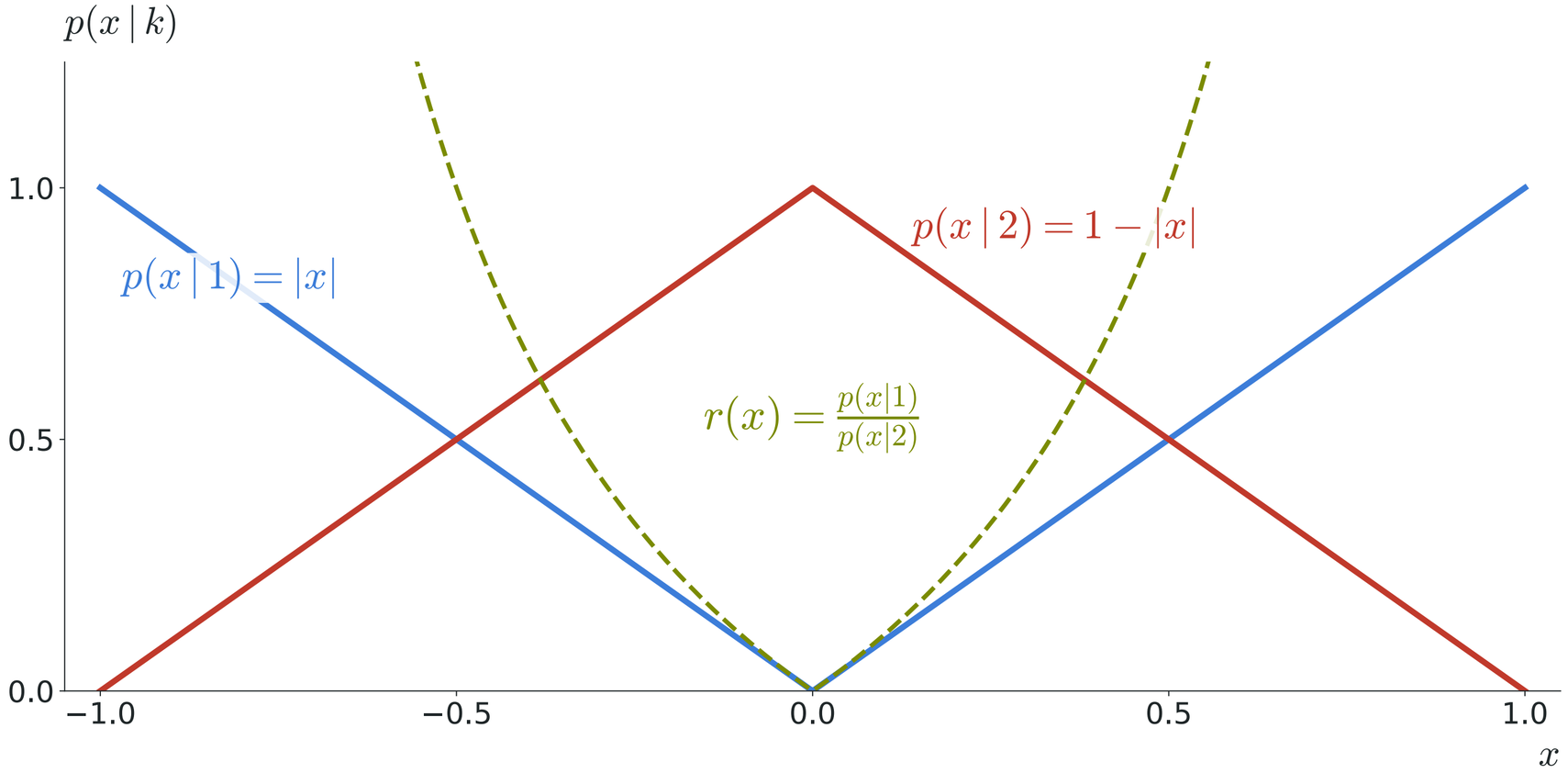
❗ For continuous x the errors change continuously, one grows and the other shrinks. The optimum is where $\epsilon(1) = \epsilon(2)$.

Example I: two triangles



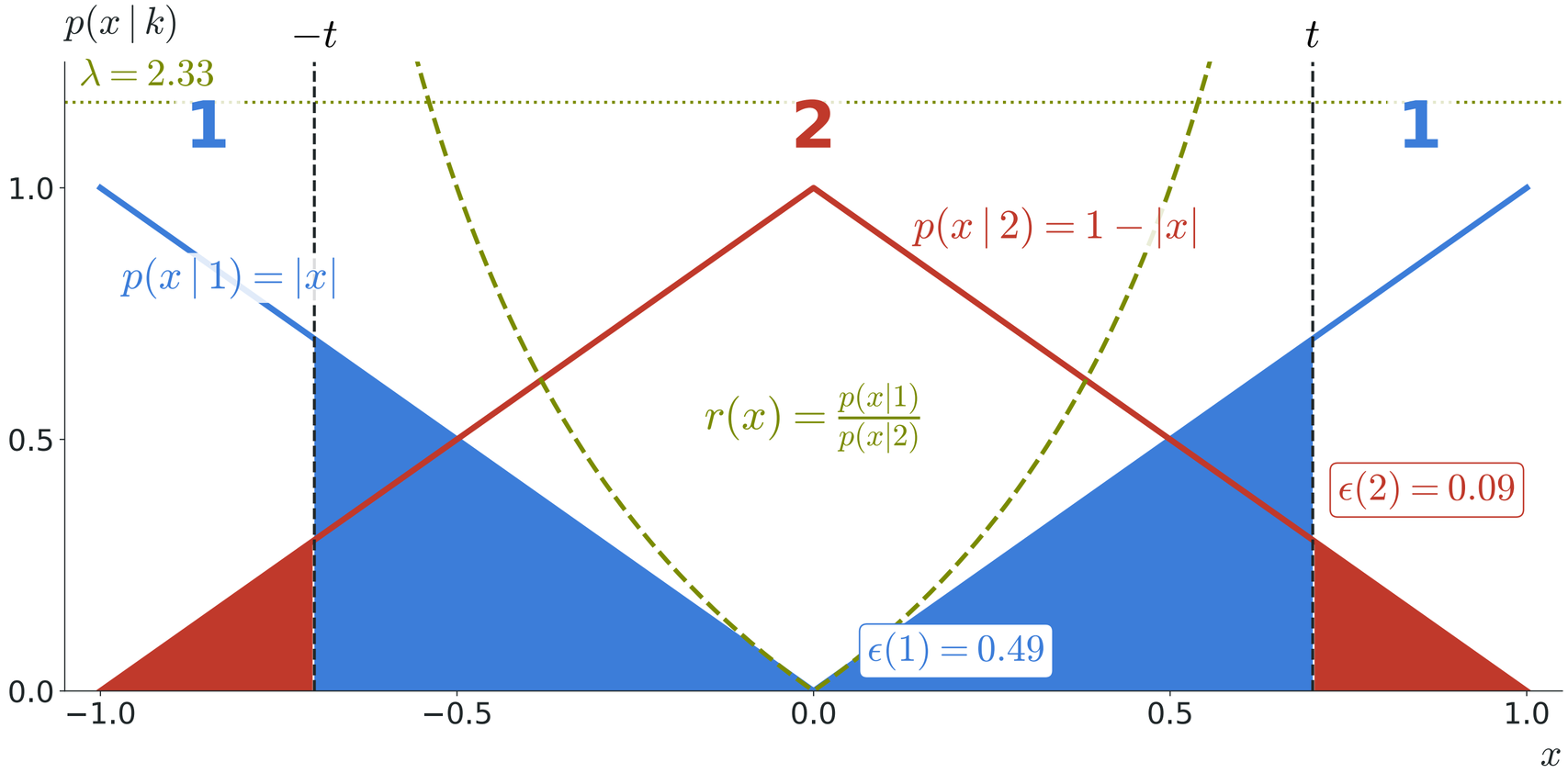
$p(x \mid 1) = |x|, \quad p(x \mid 2) = 1 - |x|, \quad x \in [-1, 1]$. Find the minimax strategy.

Example I: two triangles



Step 1: draw the likelihood ratio $r(x)$

Example I: two triangles



Step 2: a threshold λ gives the regions $1 \mid 2 \mid 1$ and the errors. Which λ is optimal?

Example I: try it

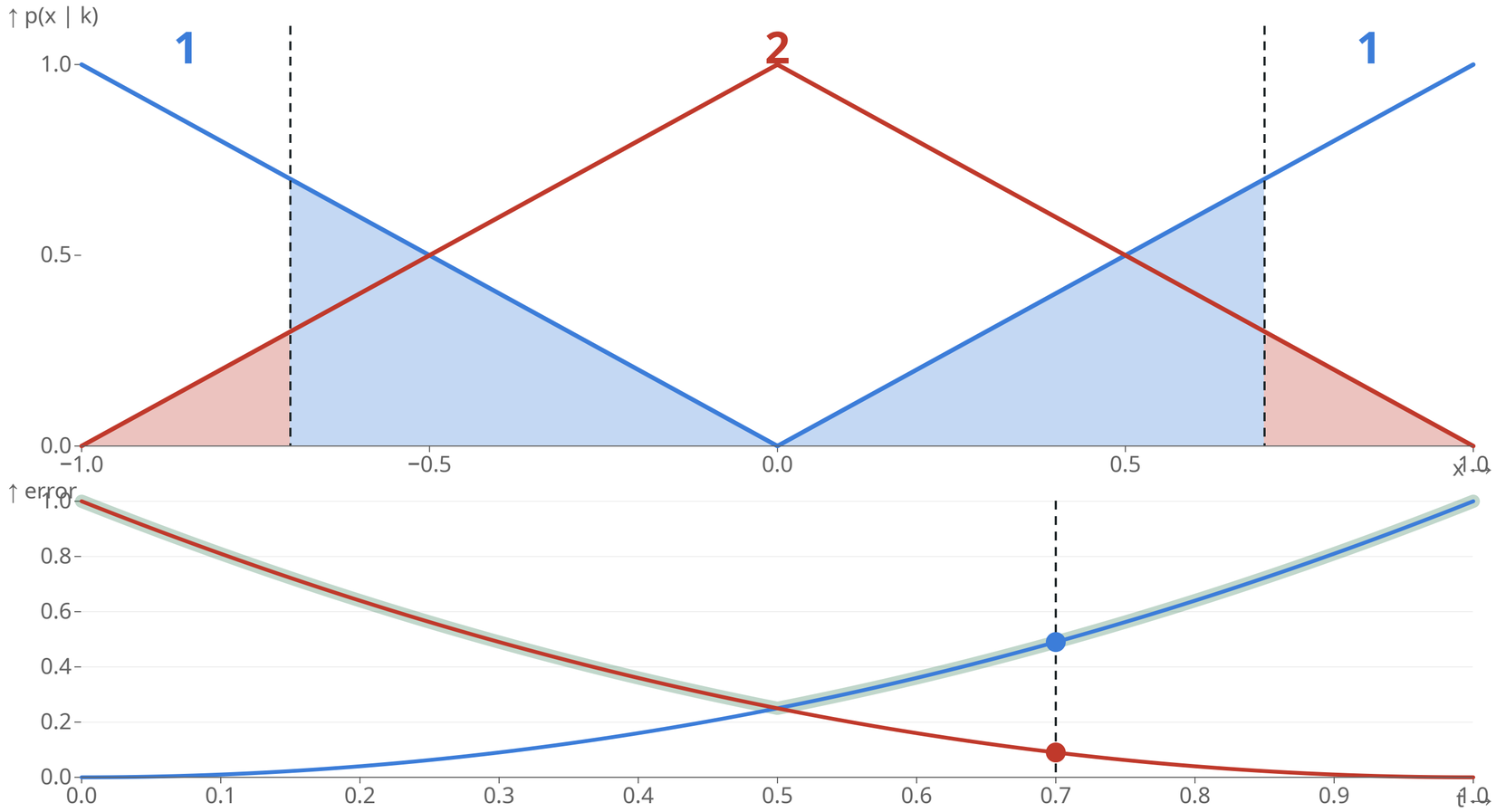
t 0.7

λ = **2.33**

ε(1) = **0.490**

ε(2) = **0.090**

max = **0.490**



Derivation: example I

By symmetry the regions are $1 \mid 2 \mid 1$ with the boundaries $\pm t$:

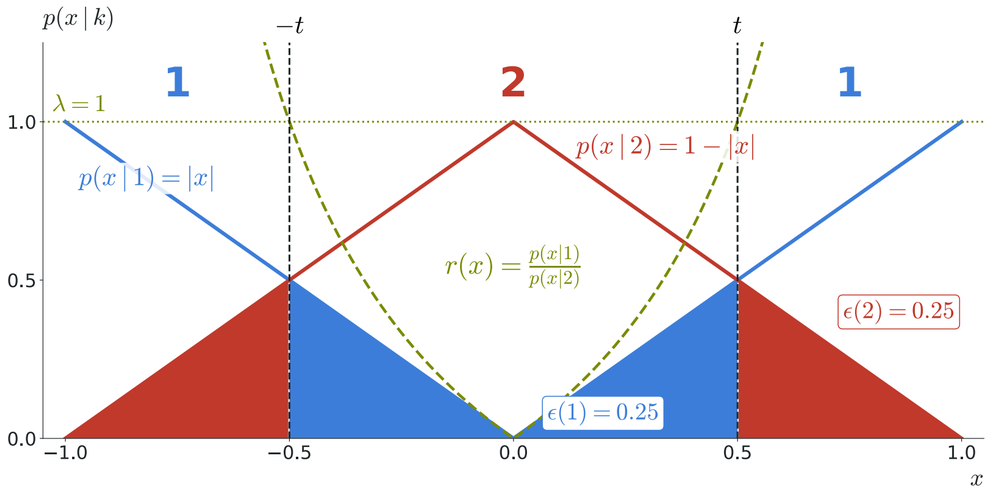
$$\epsilon(1) = \int_{-t}^{t} |x|\, dx = 2\int_{0}^{t} x\, dx = t^2$$

$$\epsilon(2) = 2\int_{t}^{1} (1-x)\, dx = (1-t)^2$$

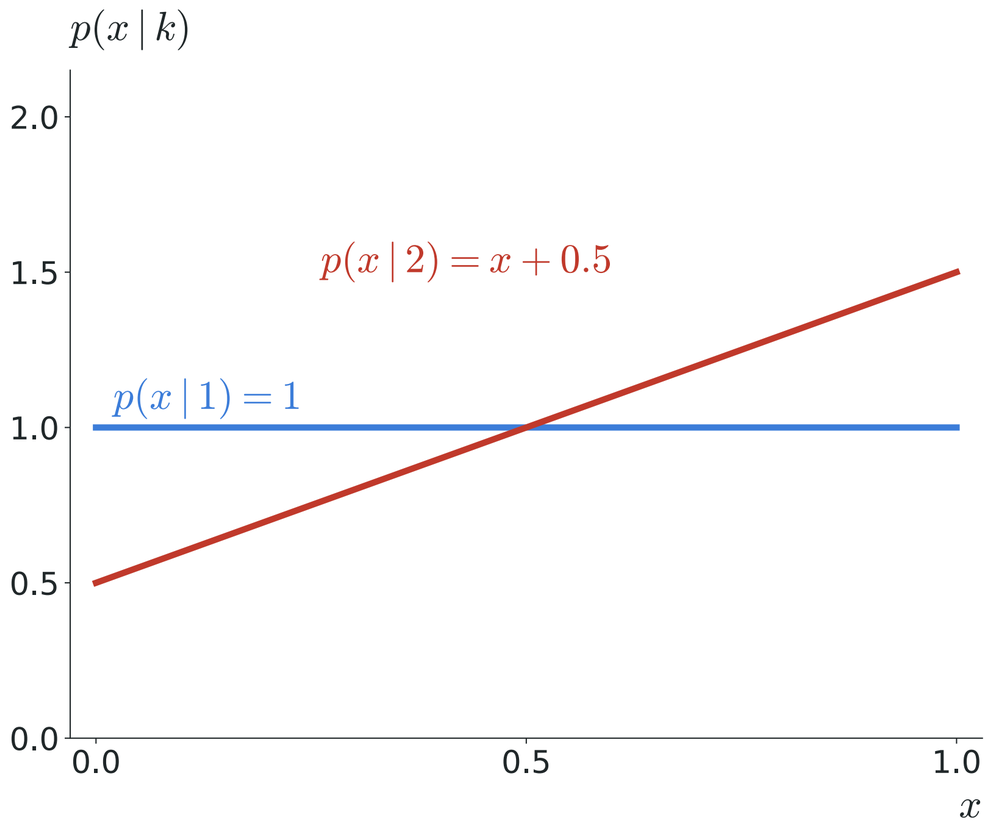
$$\epsilon(1) = \epsilon(2): \quad t^2 = (1-t)^2 \quad \text{gives} \quad t = \tfrac{1}{2}$$

$$\lambda^* = r(\tfrac{1}{2}) = 1, \qquad \epsilon(1) = \epsilon(2) = \mathbf{\frac{1}{4}}$$

$$q^*(x) = \begin{cases} 2 & \text{if } |x| \le \frac{1}{2} \\ 1 & \text{otherwise} \end{cases}$$



Example II: your turn



$$p(x \mid 1) = 1, \quad p(x \mid 2) = x + 0.5$$

on the interval $x \in (0, 1)$

? 1. What does $r(x)$ look like?

? 2. Where do we decide 1 and where 2?

? 3. Which boundary t gives $\epsilon(1) = \epsilon(2)$?

Derivation: example II

$r(x) = \frac{1}{x+0.5}$ is decreasing: we decide 1 for $x < t$.

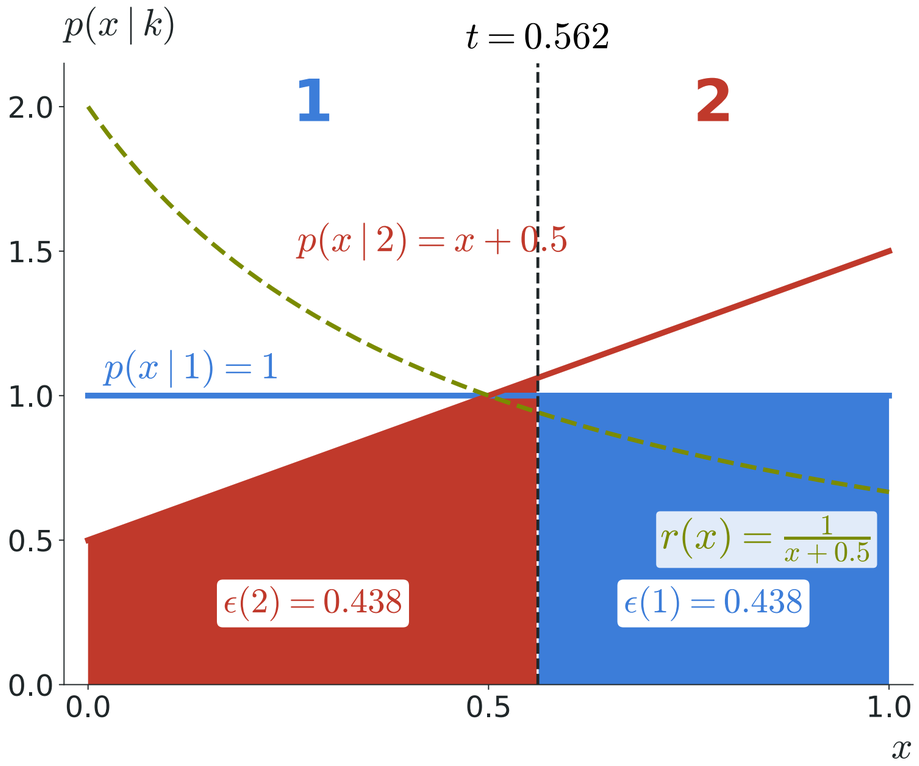
$$\epsilon(1) = \int_t^1 1\,dx = 1 - t$$

$$\epsilon(2) = \int_0^t (x + 0.5)\,dx = \tfrac{1}{2}t^2 + \tfrac{1}{2}t$$

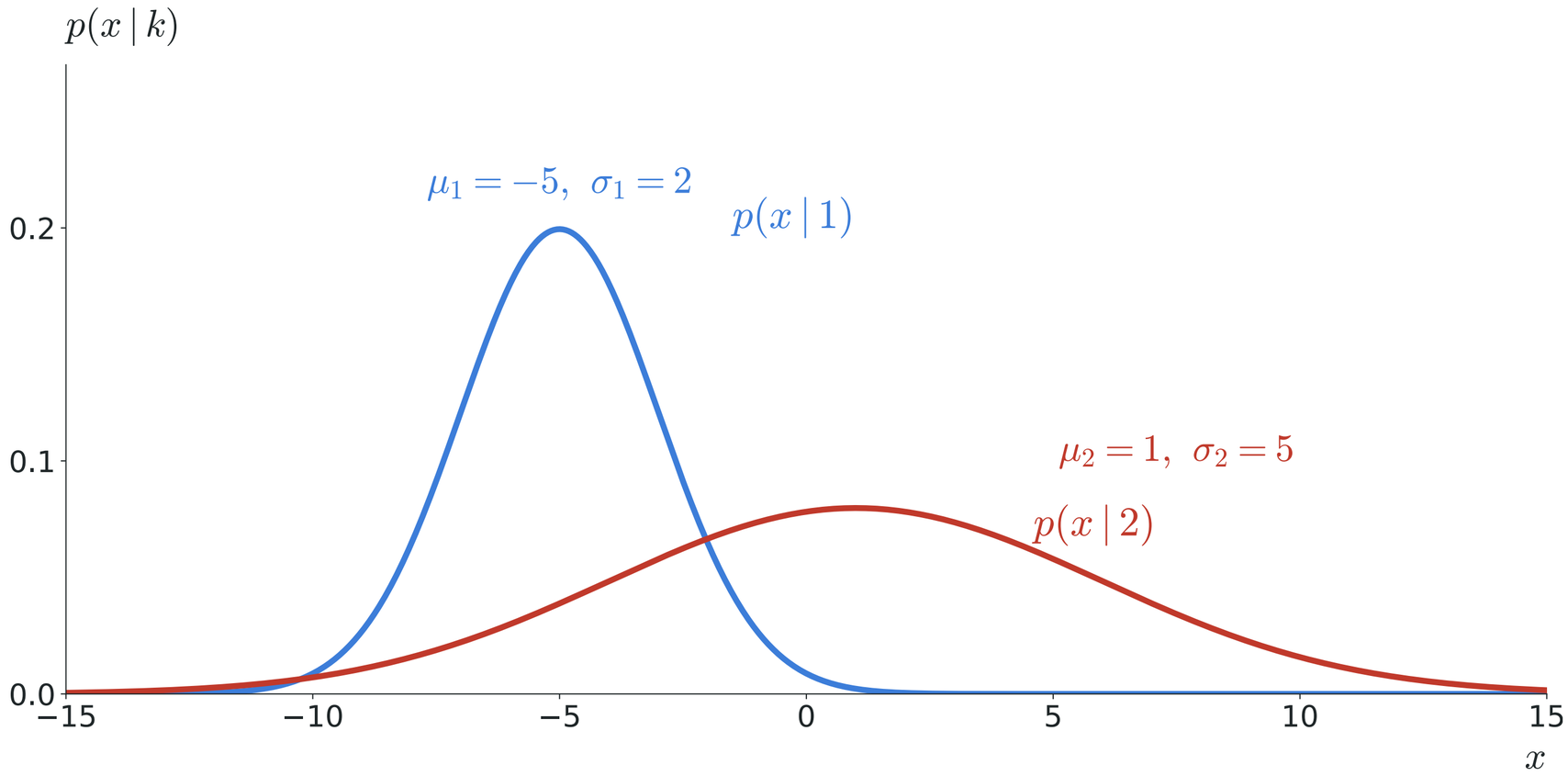
$$t^2 + 3t - 2 = 0, \qquad t = \tfrac{1}{2}\left(\sqrt{17} - 3\right) \approx 0.562$$

$$\epsilon(1) = \epsilon(2) \approx \mathbf{0.438}, \qquad \lambda^* = r(t) \approx 0.942$$

$$q^*(x) = \begin{cases} 1 & \text{if } x < 0.562 \\ 2 & \text{otherwise} \end{cases}$$

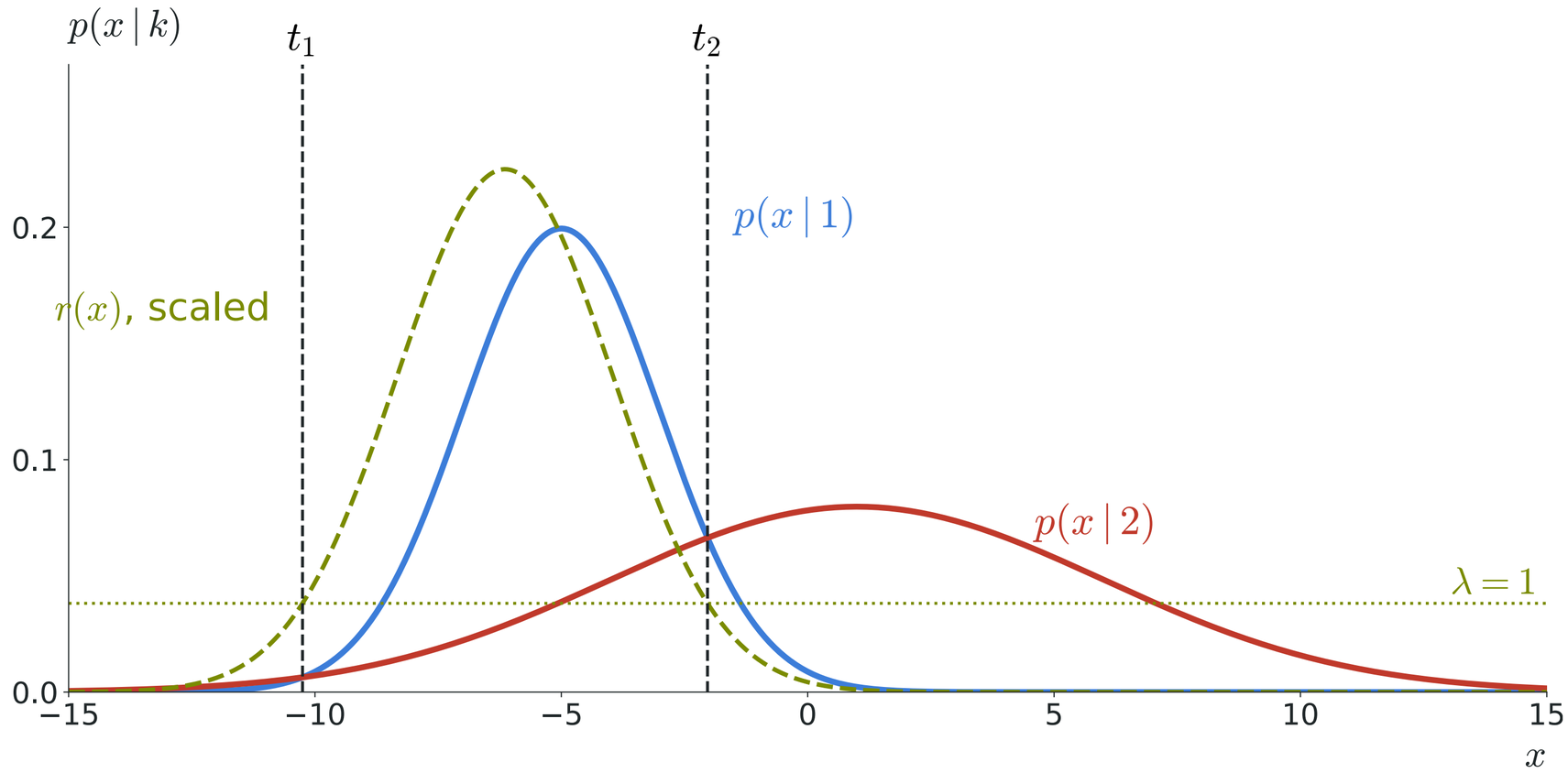


Example III: two normal distributions



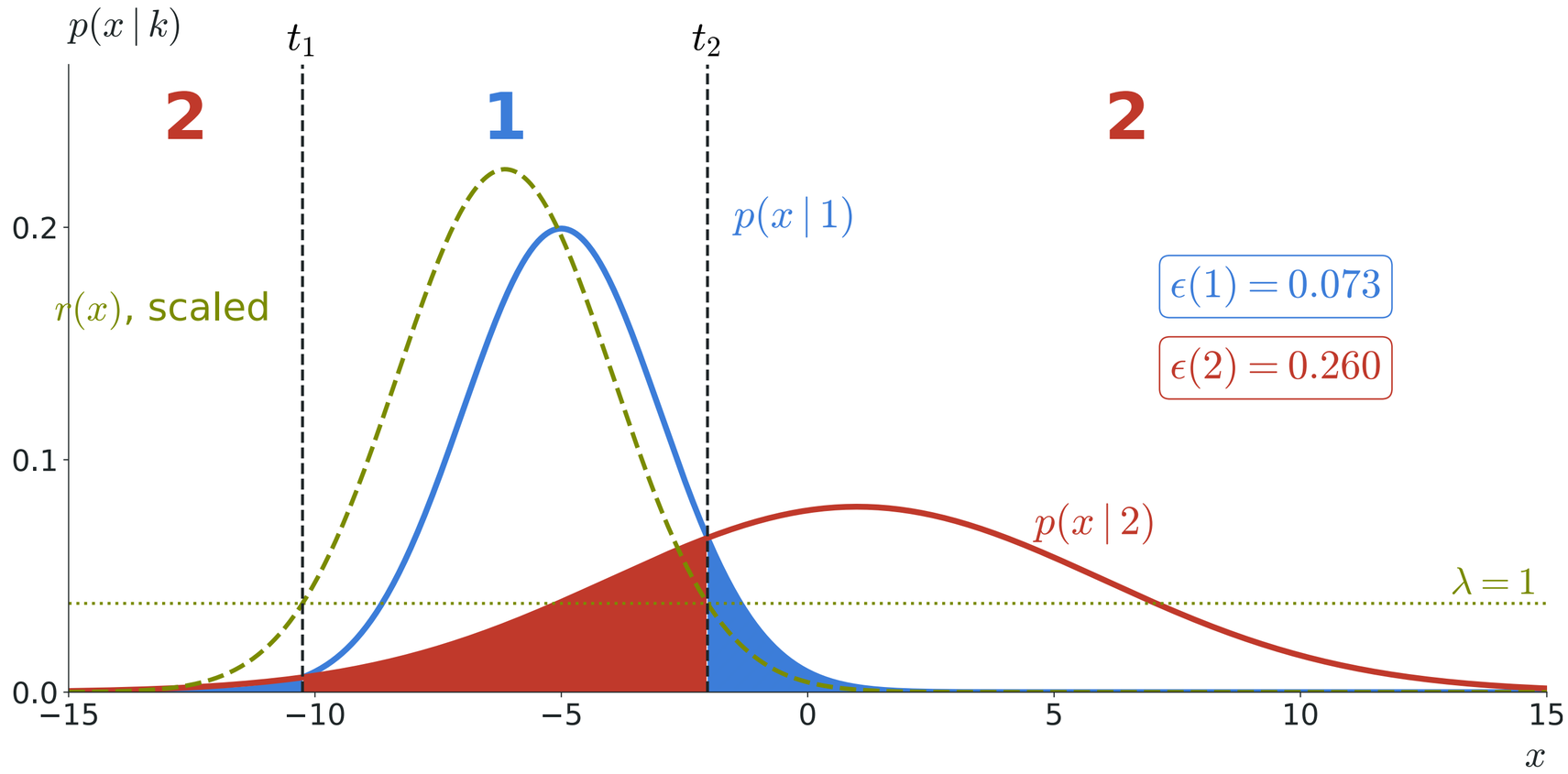
$p(x \mid k)$ normal with $\mu_1 = -5,\ \sigma_1 = 2,\ \mu_2 = 1,\ \sigma_2 = 5$. Find the minimax strategy.

Example III: two normal distributions



A threshold λ on $r(x)$ gives two boundaries t_1, t_2. How do we compute them?

Example III: two normal distributions



Regions $2 \mid 1 \mid 2$ and the errors for $\lambda = 1$. How do we get $\epsilon(1) = \epsilon(2)$?

Derivation: from λ to the boundaries

We decide 1 where $\dfrac{p(x \mid 1)}{p(x \mid 2)} \geq \lambda$:

$$\frac{\sigma_2}{\sigma_1} \exp\left(-\frac{(x-\mu_1)^2}{2{\sigma_1}^2} + \frac{(x-\mu_2)^2}{2{\sigma_2}^2} \right) \geq \lambda$$

Take the logarithm: the same inequality $ax^2 + bx + c \geq 0$ as in Lab 2

$$-\frac{(x-\mu_1)^2}{2{\sigma_1}^2} + \frac{(x-\mu_2)^2}{2{\sigma_2}^2} \geq \ln\left(\frac{\sigma_1}{\sigma_2} \cdot \lambda \right) =: L$$

$a = {\sigma_1}^2 - {\sigma_2}^2$ $\quad b = 2({\sigma_2}^2\mu_1 - {\sigma_1}^2\mu_2)$ $\quad c = {\sigma_1}^2{\mu_2}^2 - {\sigma_2}^2{\mu_1}^2 - 2{\sigma_1}^2{\sigma_2}^2 L$

! Lab 2 had $\dfrac{p(2)}{p(1)}$ in place of λ: this is the **Bayesian strategy** for $p(1) = \dfrac{1}{1+\lambda}$

Derivation: the errors of the strategy

For $\sigma_1 < \sigma_2$ the regions are $2 \mid 1 \mid 2$ with the boundaries $t_1 < t_2$.

With the distribution functions F_1, F_2 (`norm.cdf`):

$$\epsilon(1) = \int_{-\infty}^{t_1} p(x \mid 1)\, dx + \int_{t_2}^{\infty} p(x \mid 1)\, dx = 1 - \big(F_1(t_2) - F_1(t_1)\big)$$

$$\epsilon(2) = \int_{t_1}^{t_2} p(x \mid 2)\, dx = F_2(t_2) - F_2(t_1)$$

! $\epsilon(1) = \epsilon(2)$ has **no closed-form solution** for normal distributions. We have to search for λ numerically.

Minimax and Bayes

The risk of a fixed strategy

We fix a strategy q. How does its risk change when the prior changes?

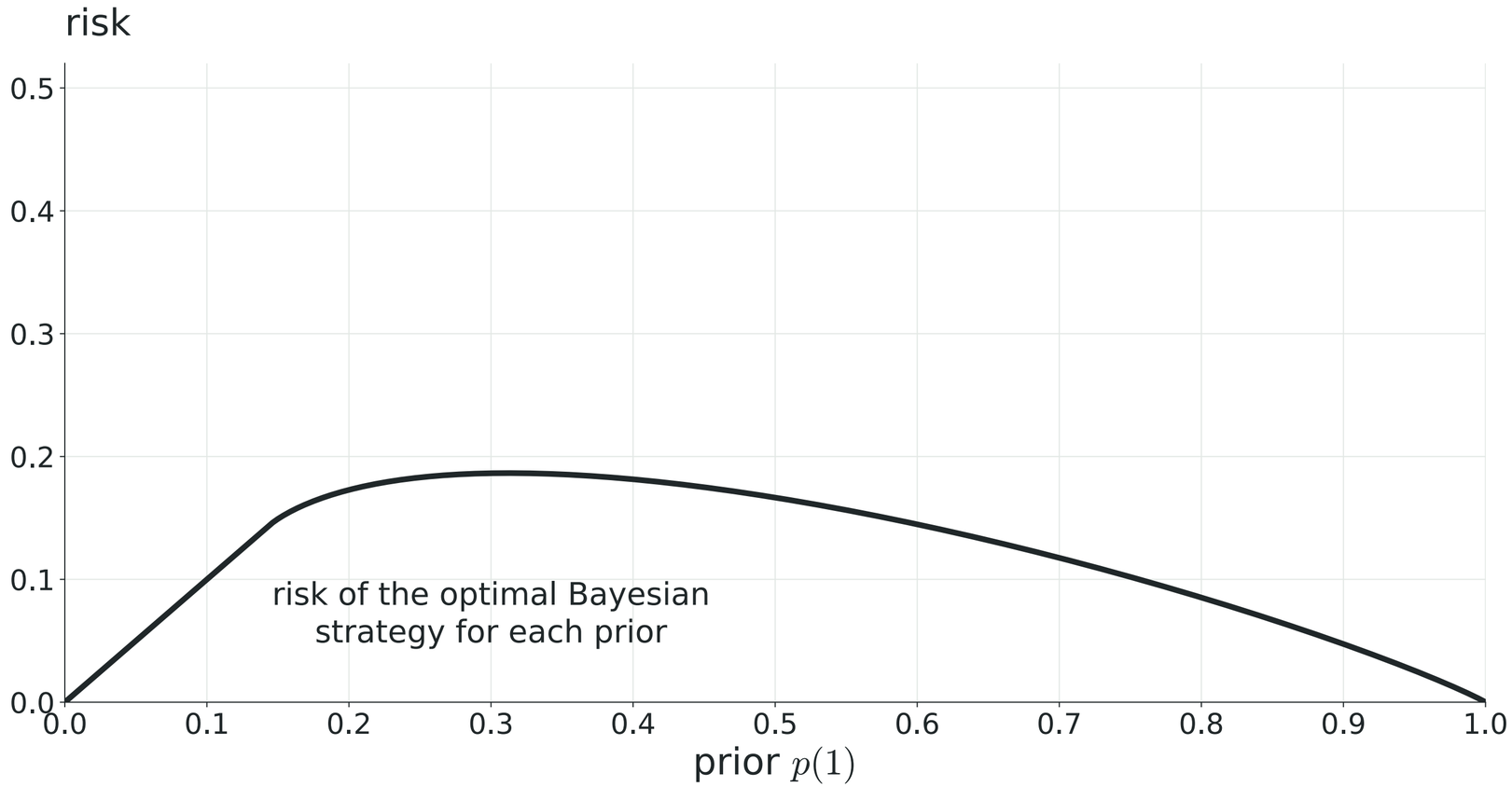
$$R(q) = \sum_k \int_{x:\, q(x)\neq k} p_{XK}(x, k)\, dx = \sum_k p(k) \underbrace{\int_{x:\, q(x)\neq k} p(x \mid k)\, dx}_{\epsilon(k)}$$

For two classes, with $p(2) = 1 - p(1)$:

$$R(q) = p(1)\,\epsilon(1) + \big(1 - p(1)\big)\,\epsilon(2) = p(1)\,\big(\epsilon(1) - \epsilon(2)\big) + \epsilon(2)$$

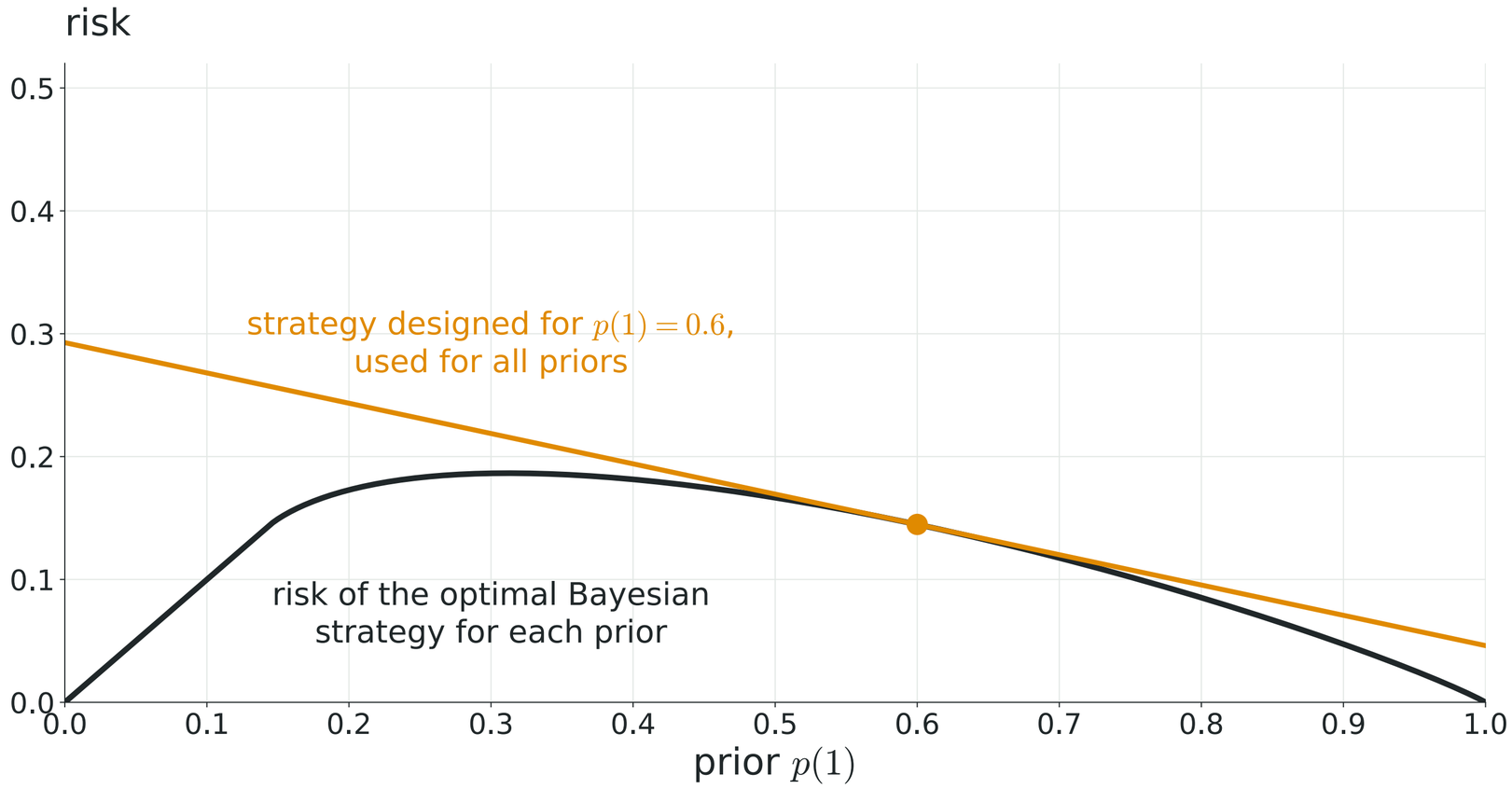
Linear in $p(1)$, from $\epsilon(2)$ at $p(1) = 0$ to $\epsilon(1)$ at $p(1) = 1$. Worst case: $\max(\epsilon(1), \epsilon(2))$

Minimax as the worst-case Bayes



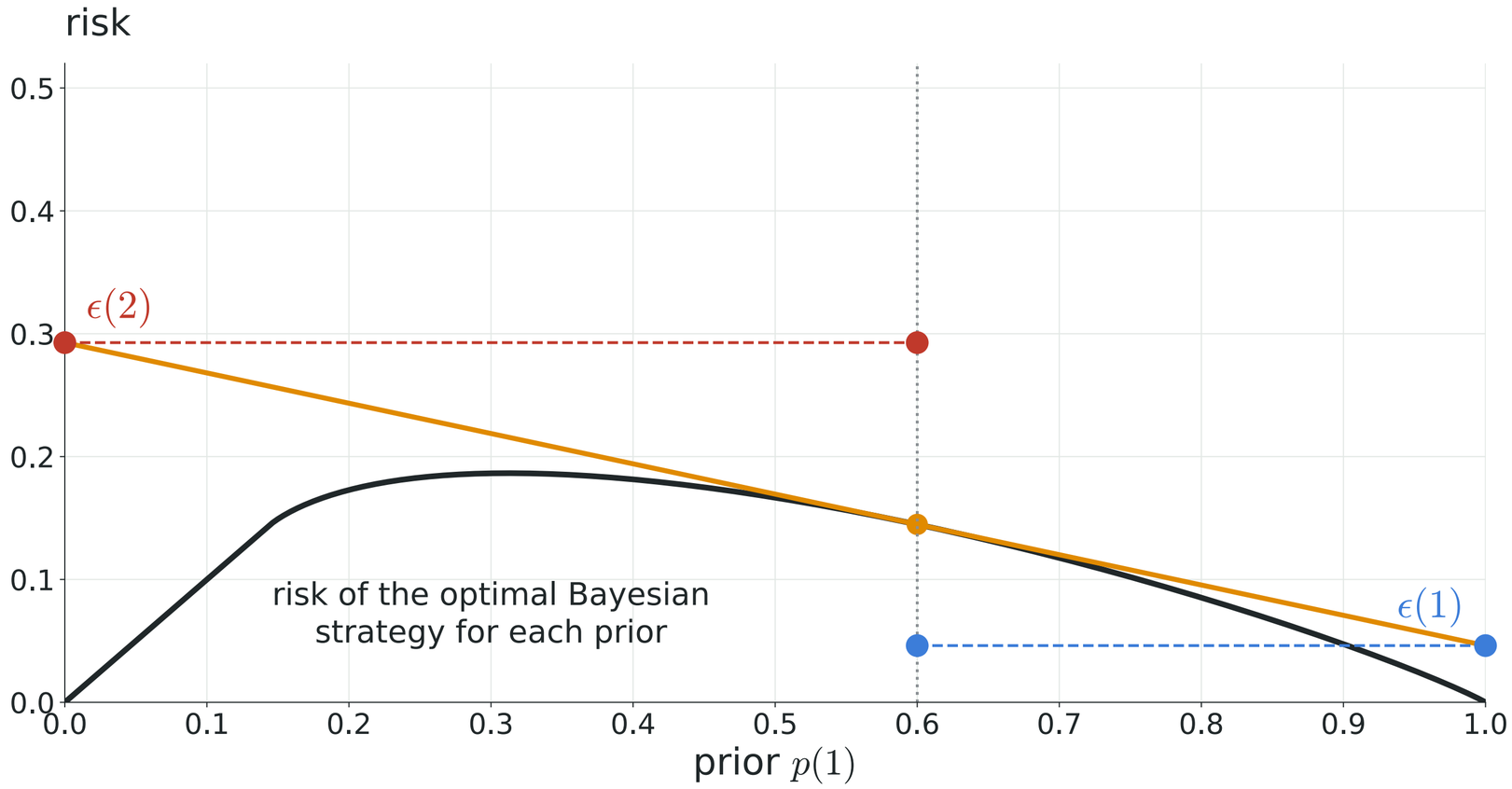
For each prior $p(1)$: the risk of **its own** optimal Bayesian strategy (example III)

Minimax as the worst-case Bayes



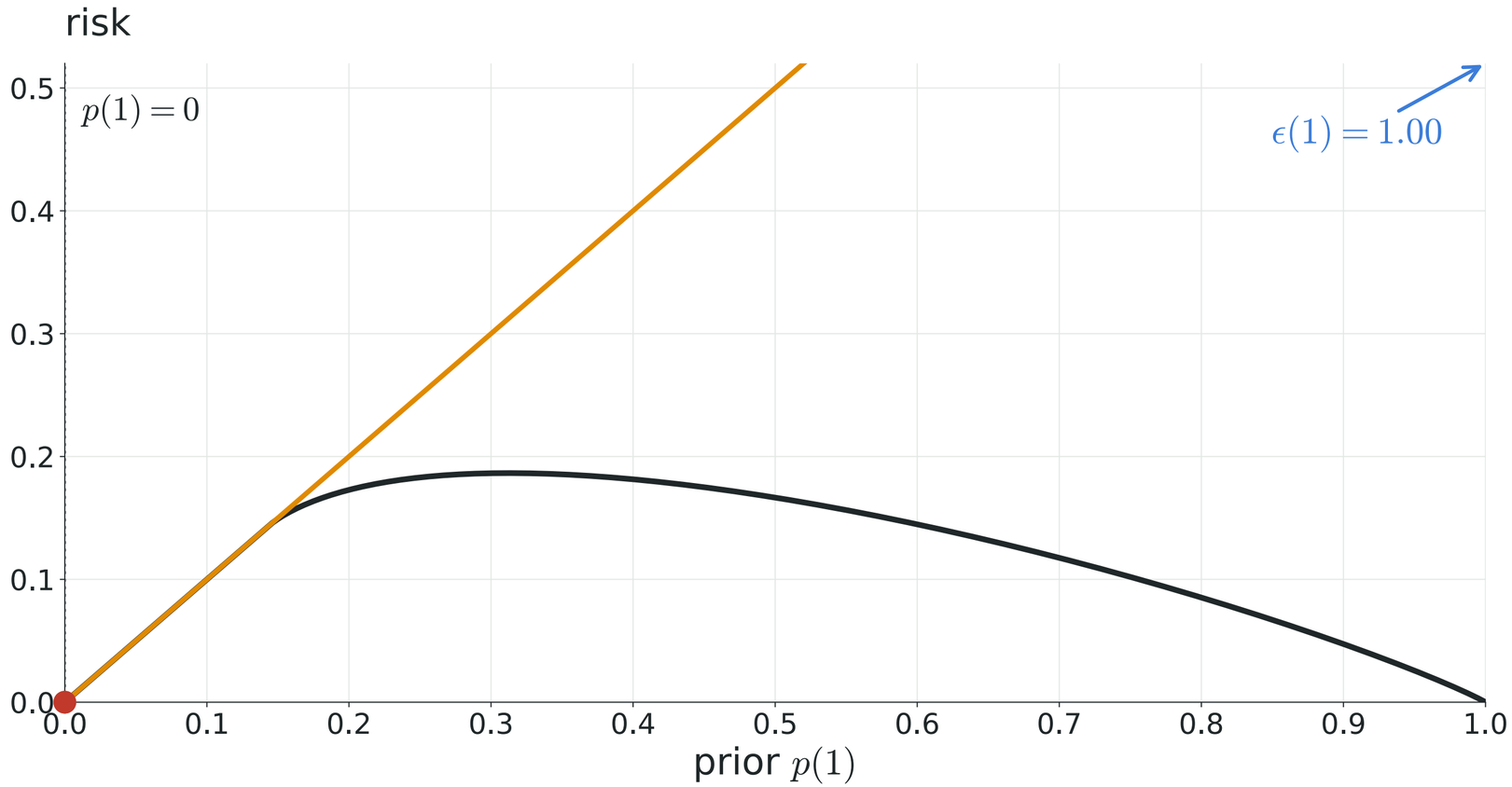
The strategy designed for $p(1) = 0.6$, used everywhere: a line touching the curve

Minimax as the worst-case Bayes



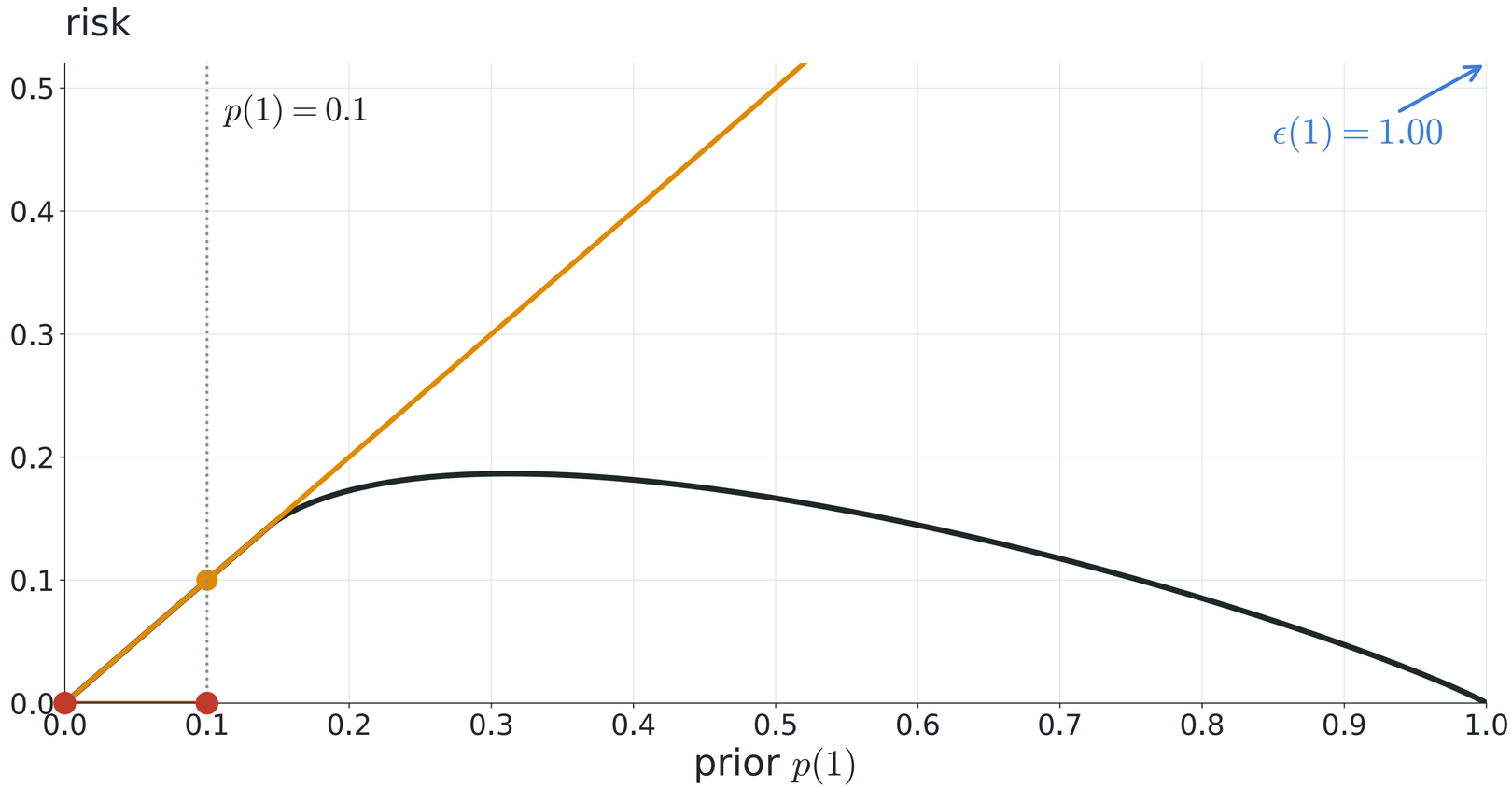
Its ends are $\epsilon(2)$ (left) and $\epsilon(1)$ (right). We mark both above its prior

Minimax as the worst-case Bayes



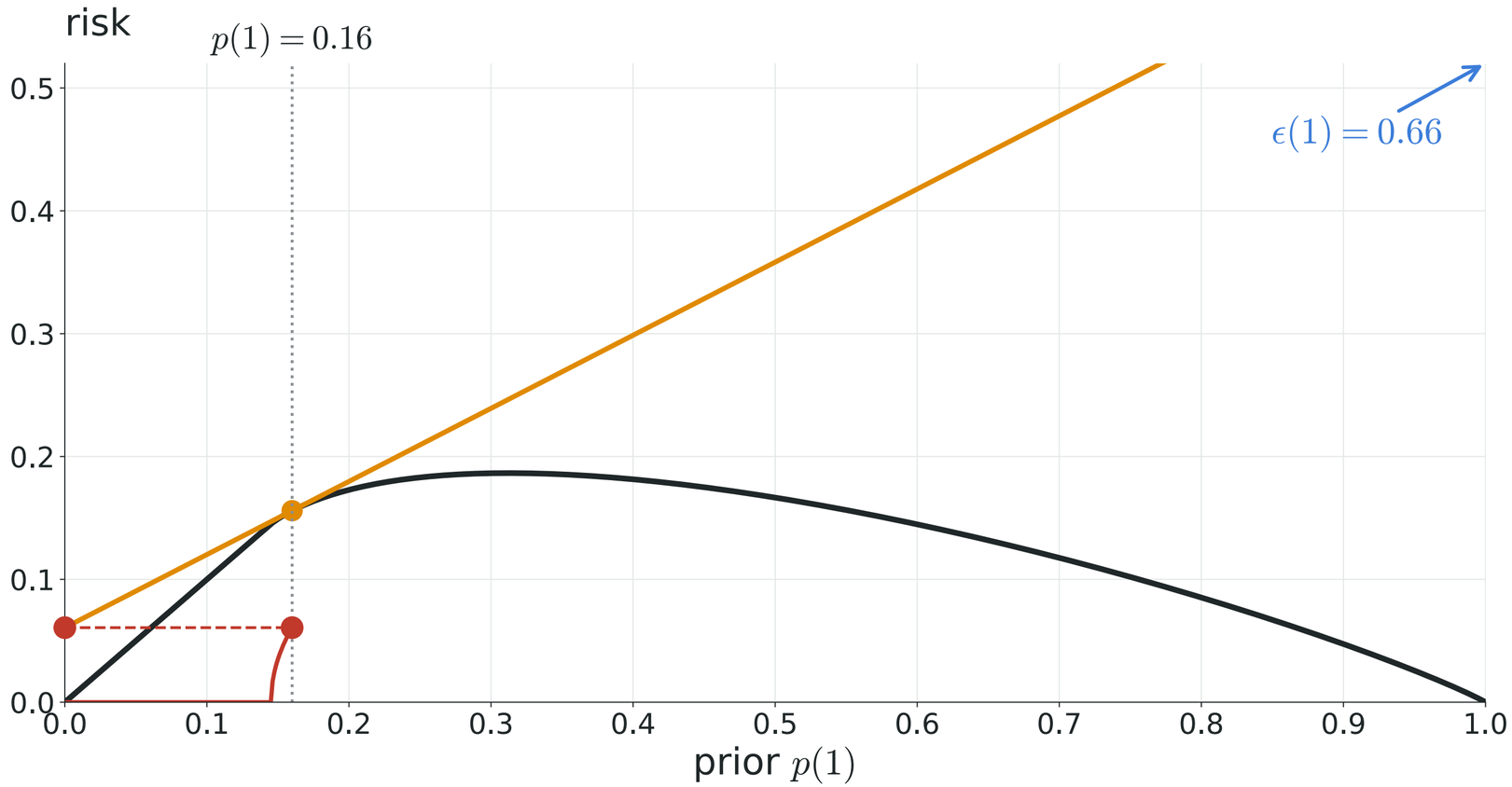
The same for the strategies of all priors from 0 to 1: the points draw $\epsilon(1)$ and $\epsilon(2)$

Minimax as the worst-case Bayes



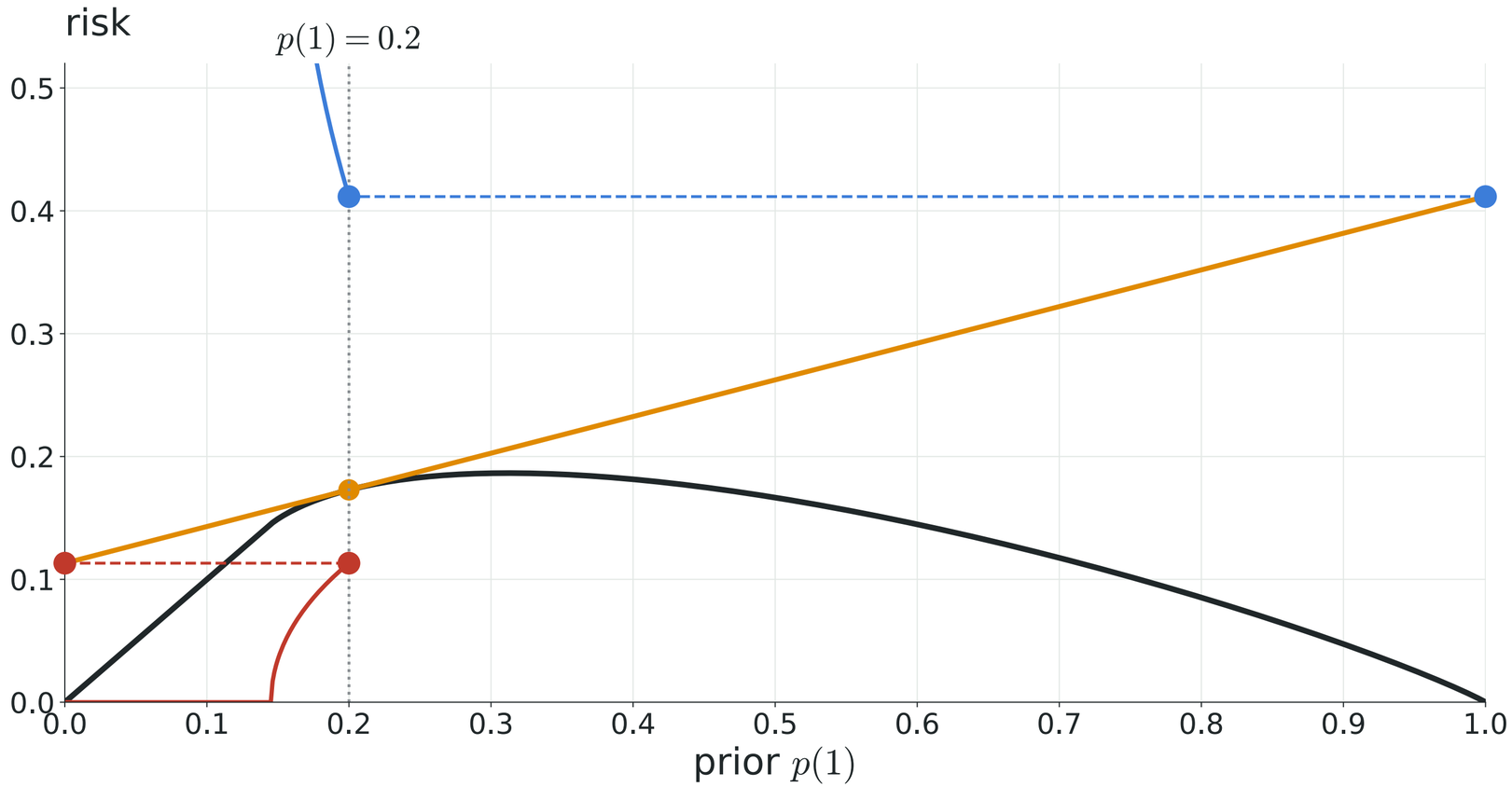
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Minimax as the worst-case Bayes



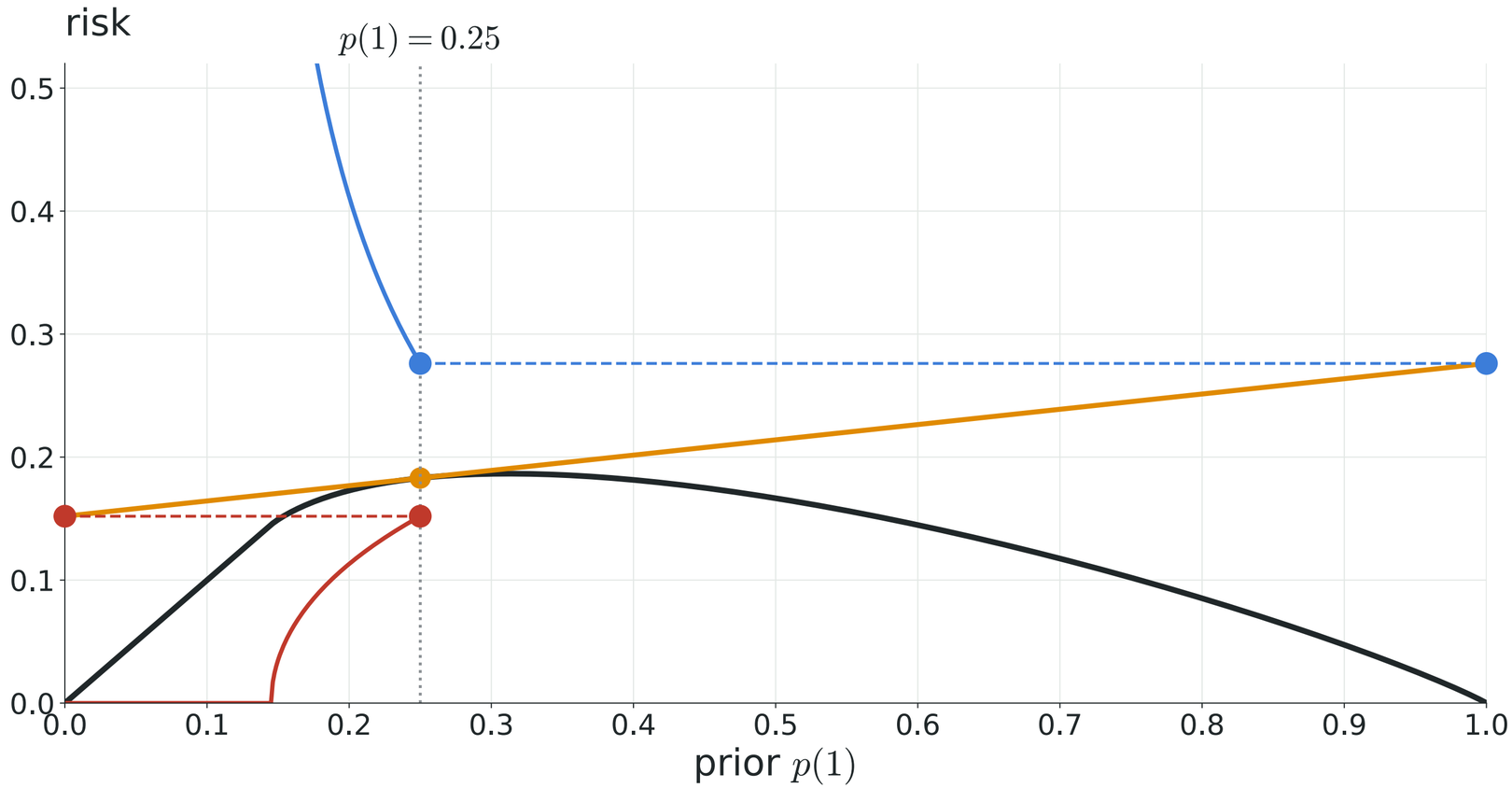
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Minimax as the worst-case Bayes



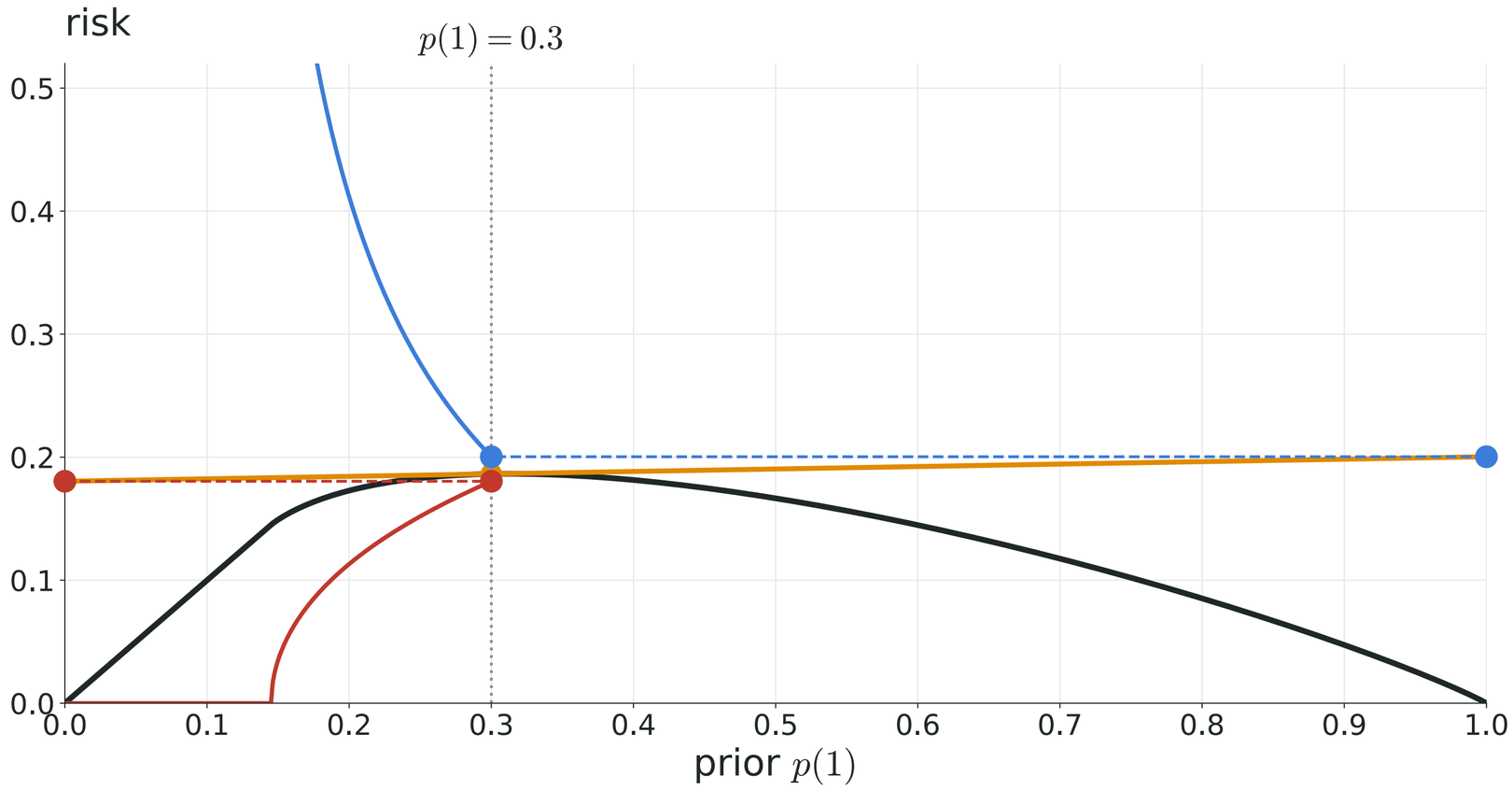
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Minimax as the worst-case Bayes



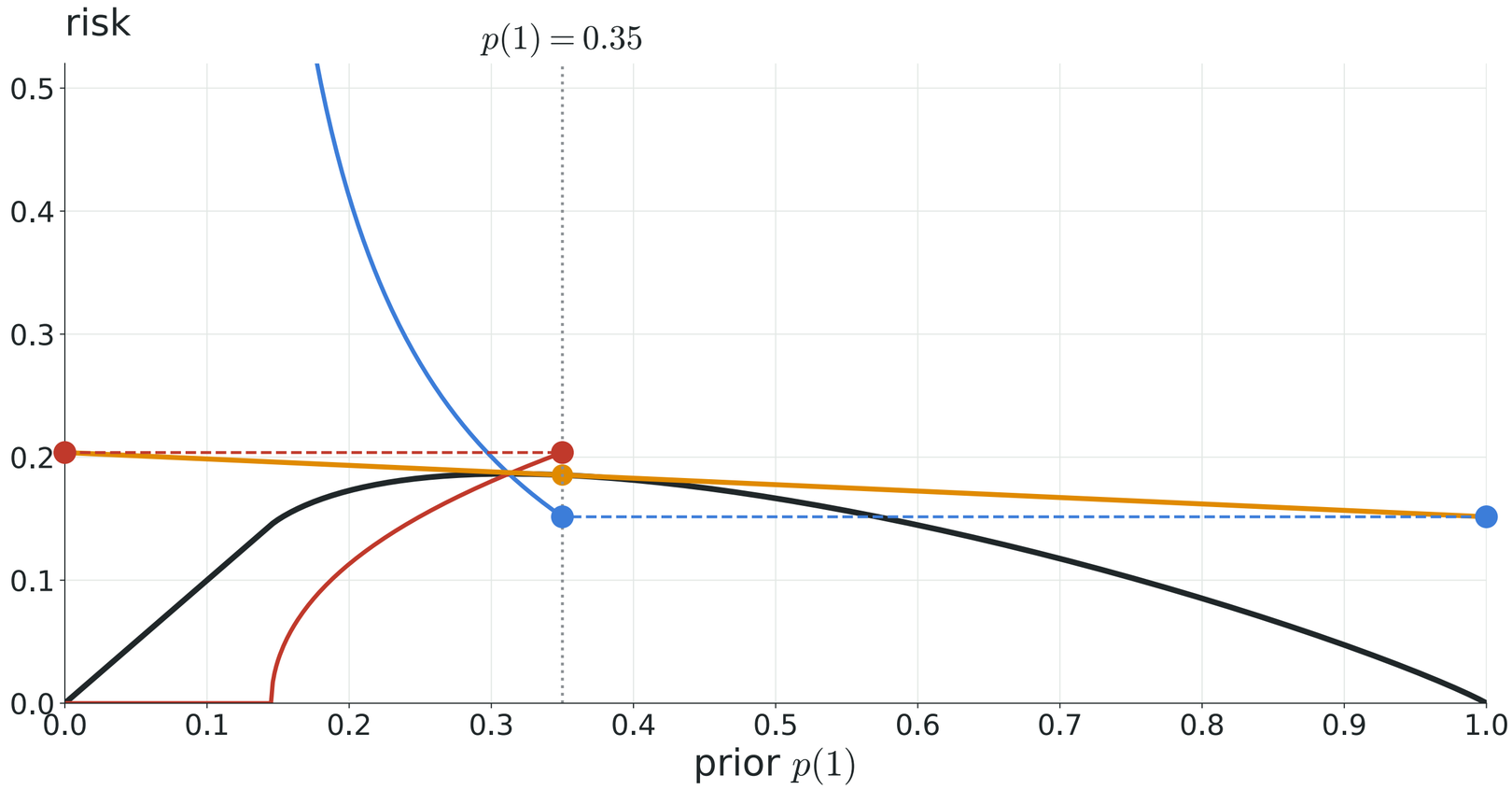
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Minimax as the worst-case Bayes



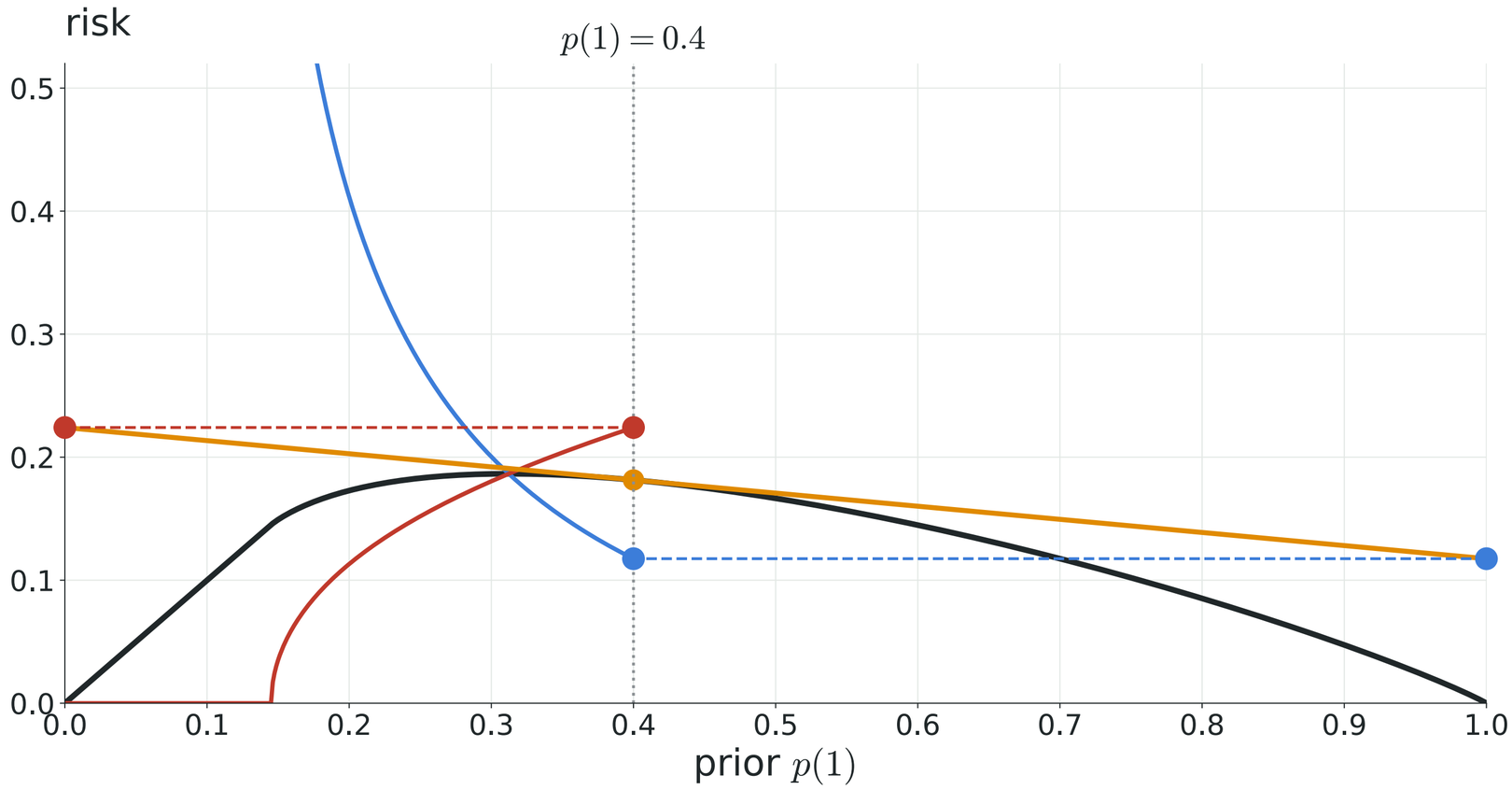
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Minimax as the worst-case Bayes



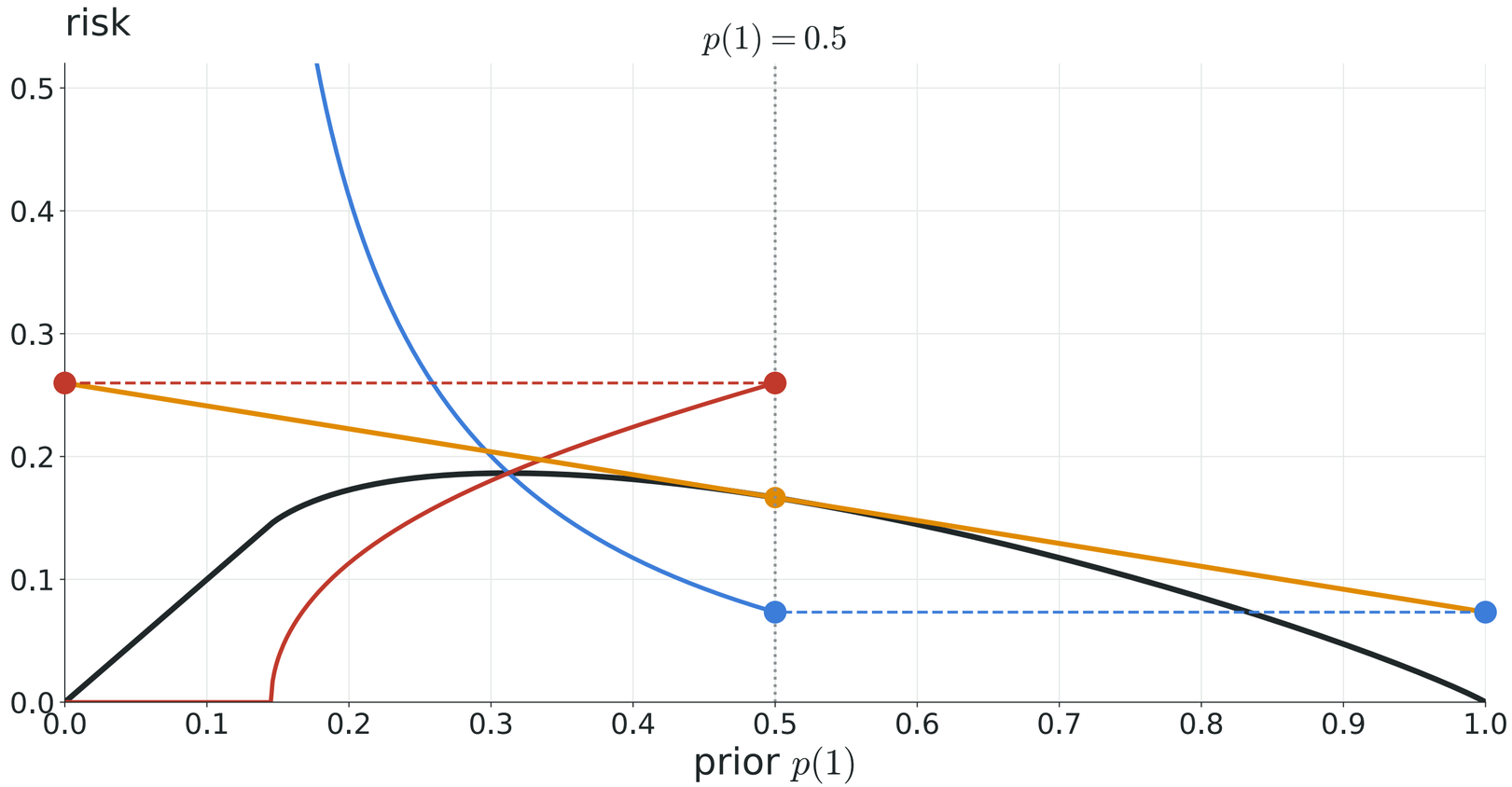
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Minimax as the worst-case Bayes



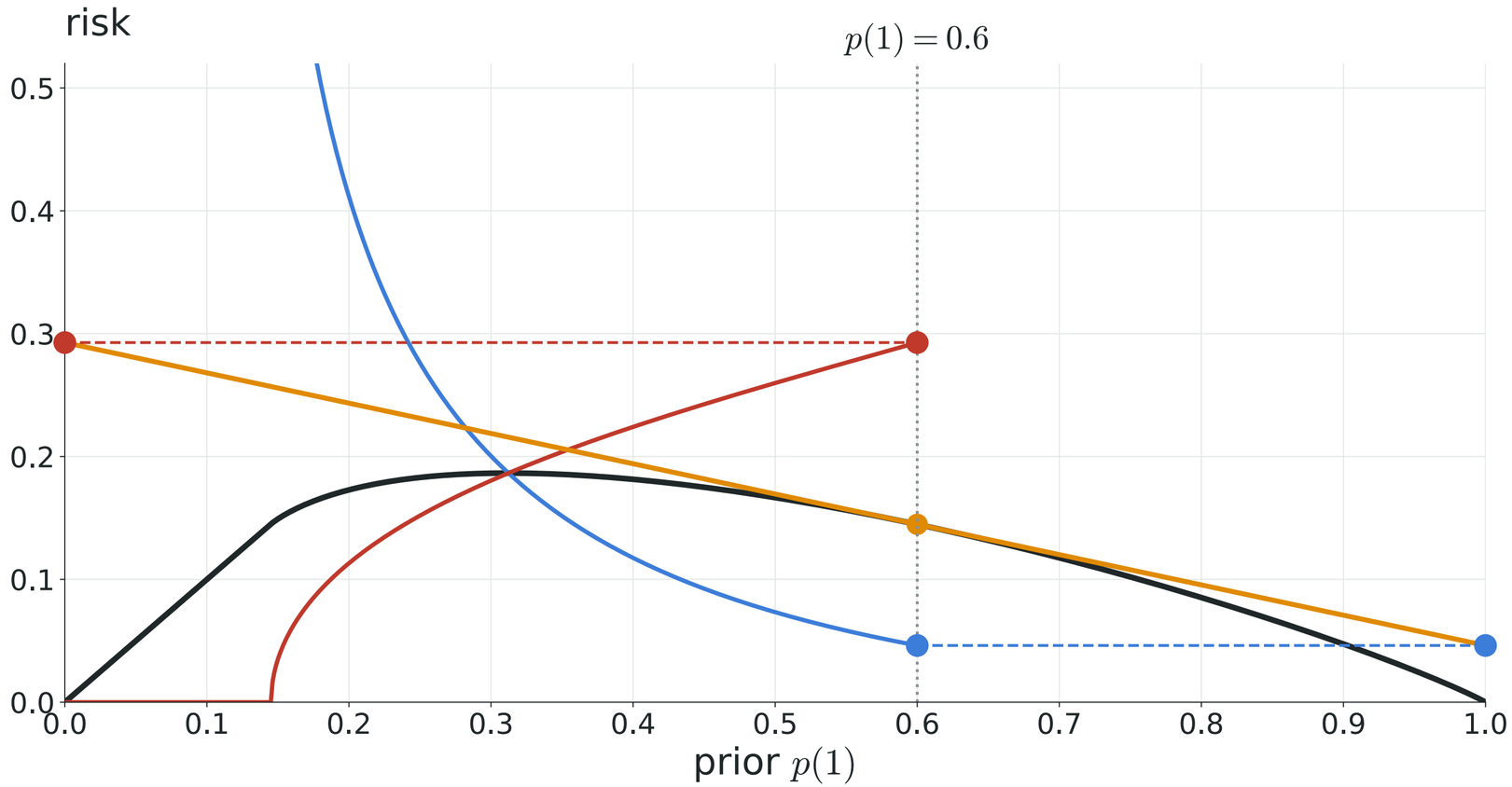
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Minimax as the worst-case Bayes



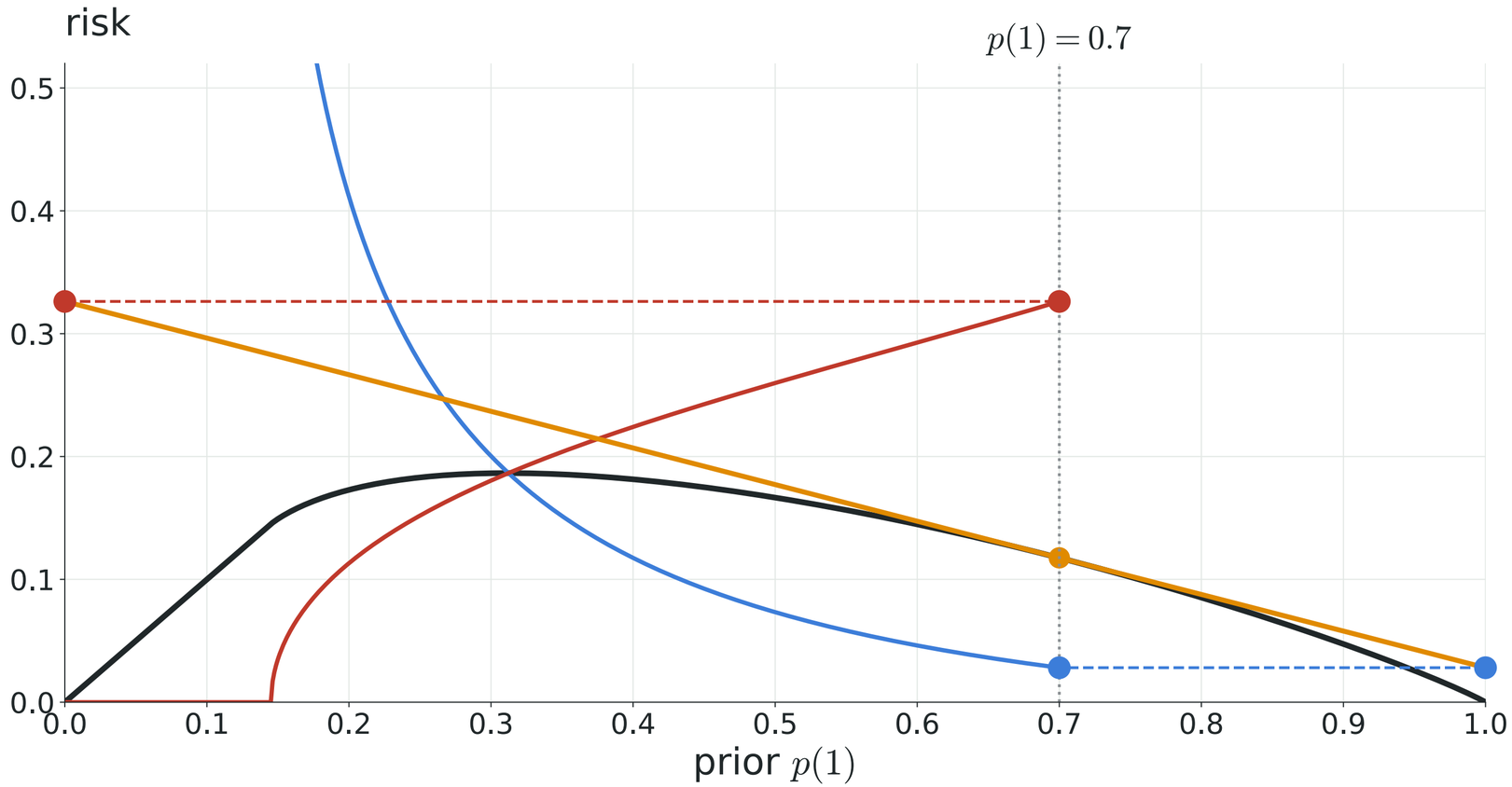
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Minimax as the worst-case Bayes



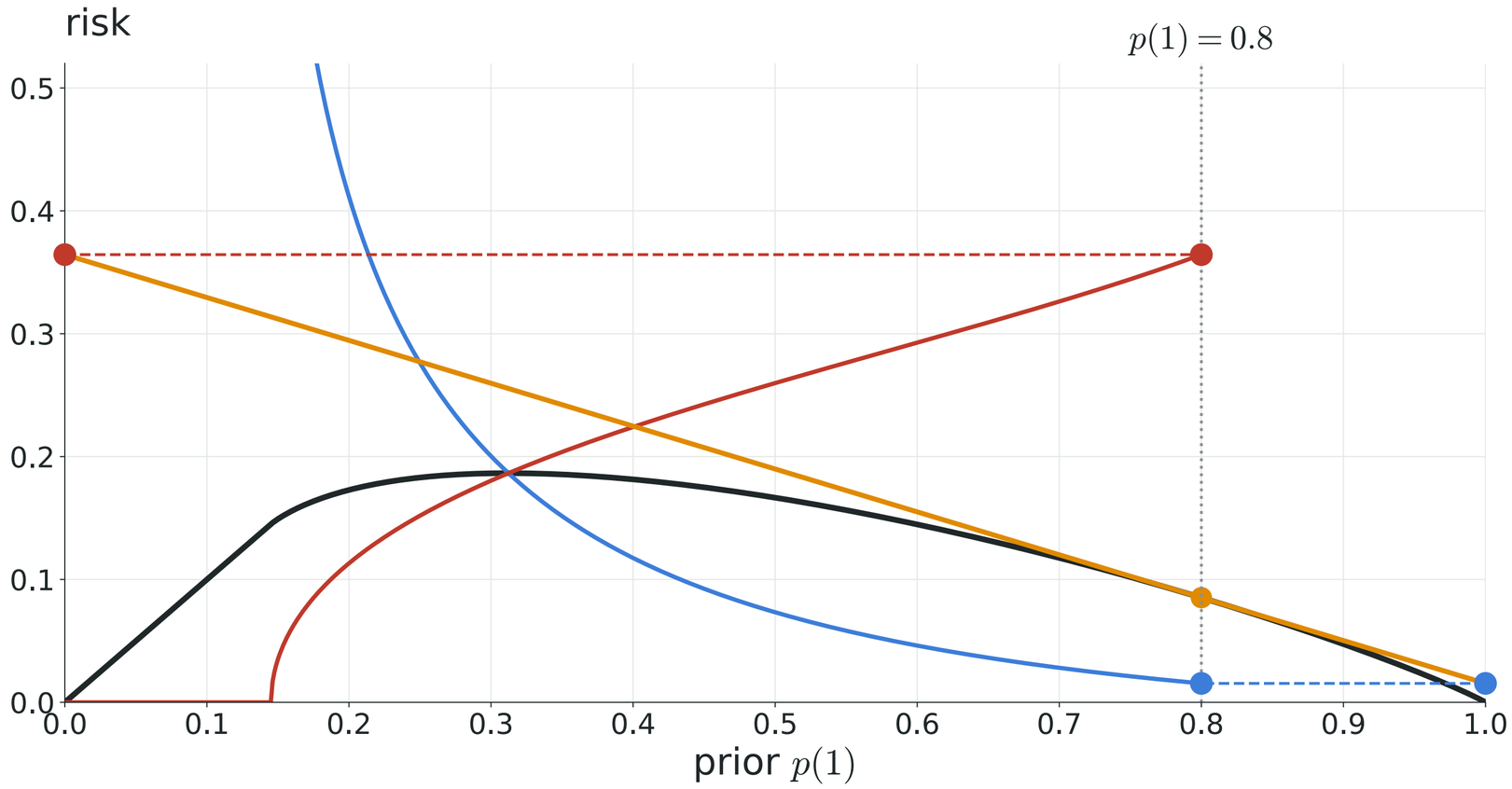
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Minimax as the worst-case Bayes



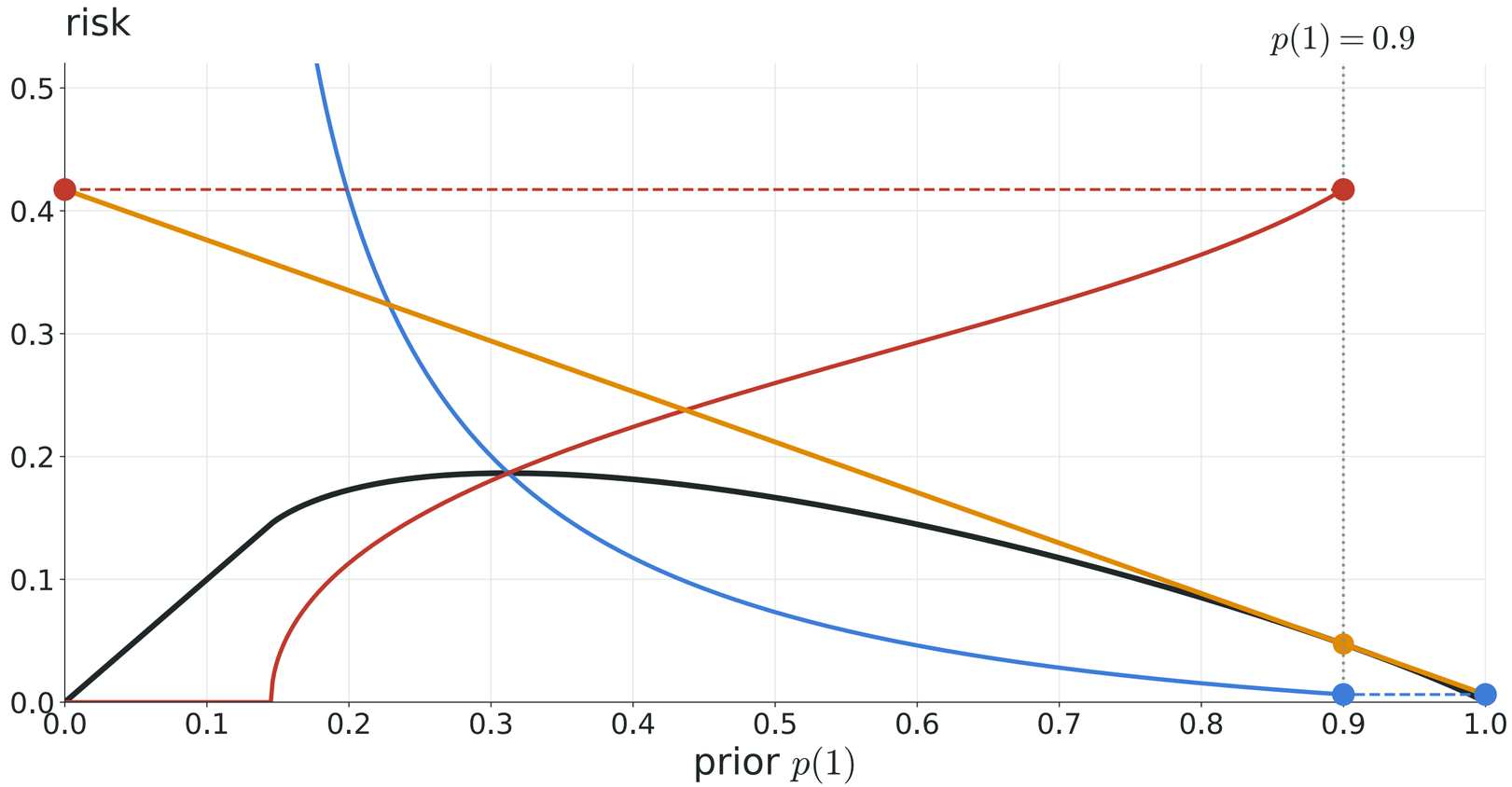
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Minimax as the worst-case Bayes



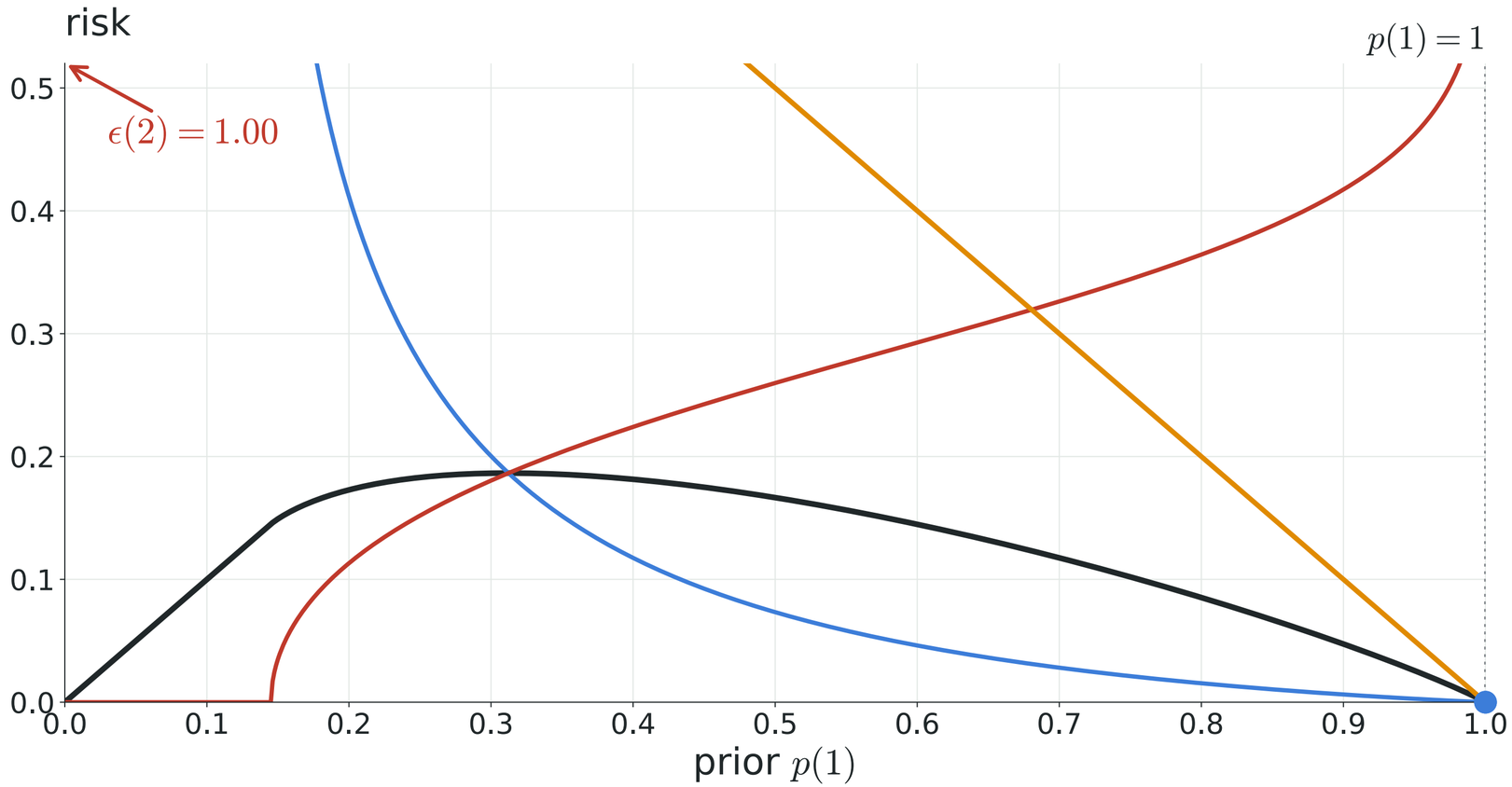
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Minimax as the worst-case Bayes



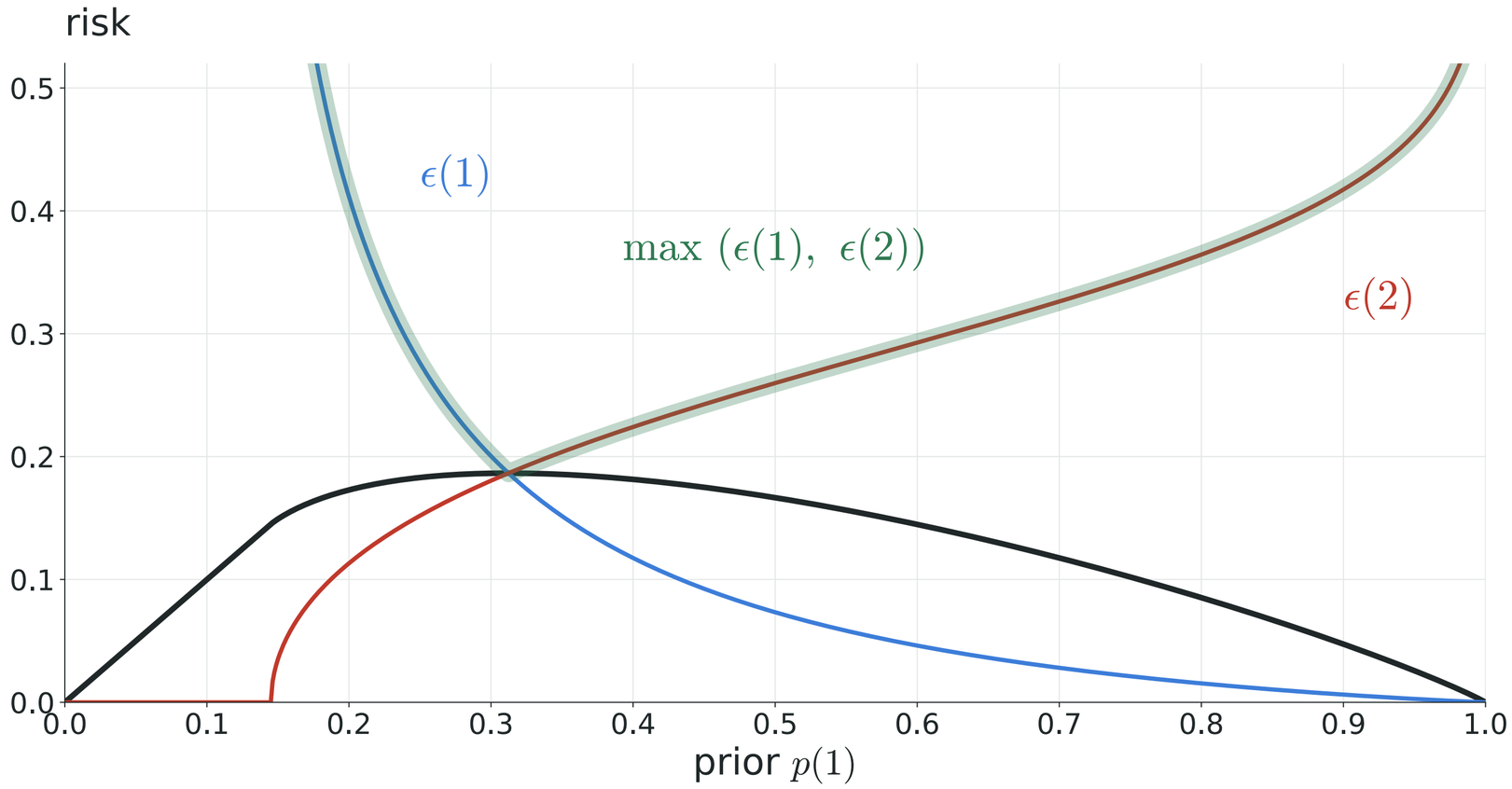
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Minimax as the worst-case Bayes



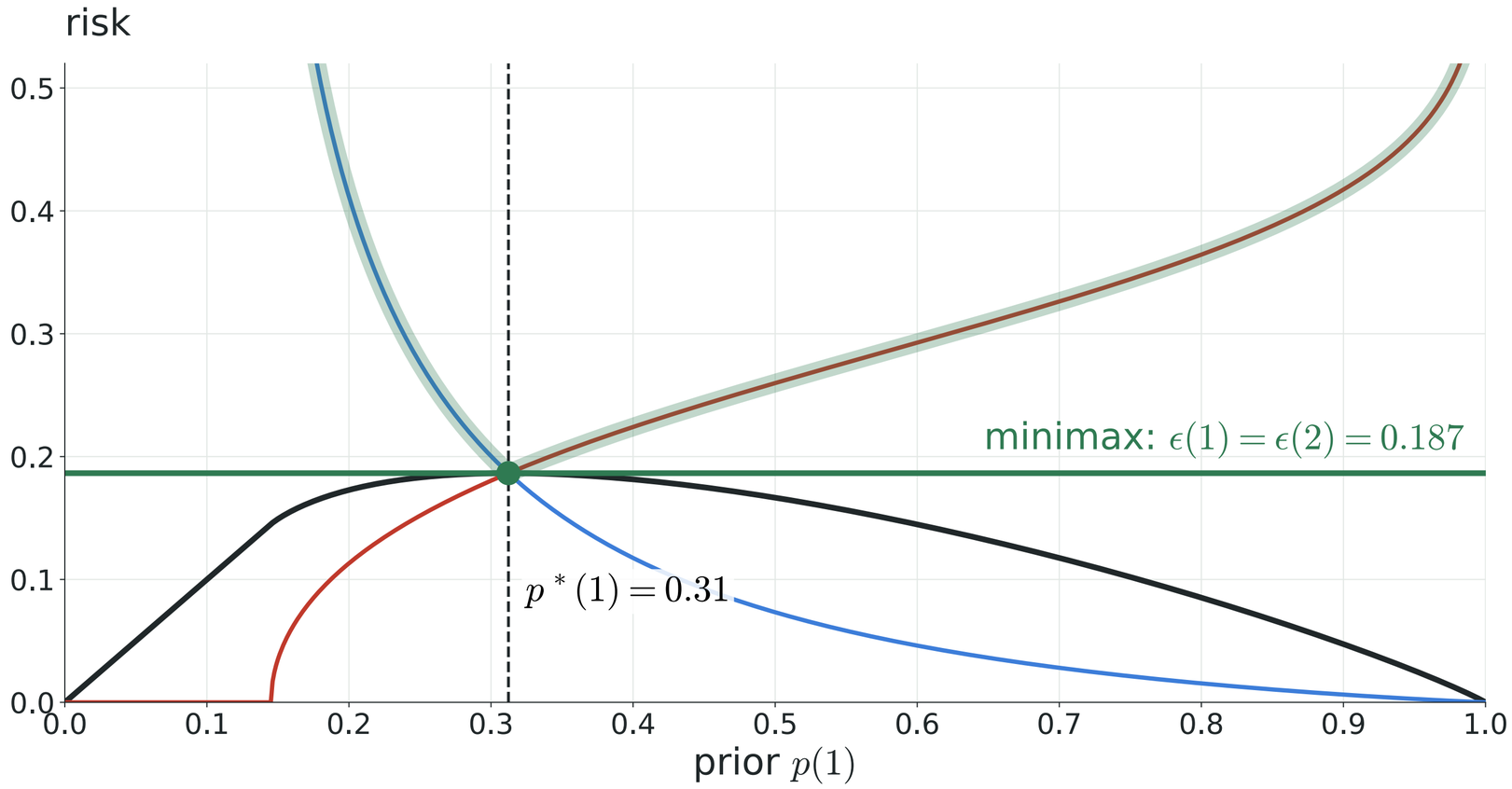
The same for the strategies of all priors from 0 to 1: the points draw $\epsilon(1)$ and $\epsilon(2)$

Minimax as the worst-case Bayes



The worst case of every Bayesian strategy: $\max(\epsilon(1), \epsilon(2))$

Minimax as the worst-case Bayes



Minimax: the Bayesian strategy for $p^*(1)$, its line is horizontal, $\epsilon(1) = \epsilon(2)$

Minimax and Bayes: try it

p(1) 0.6

λ = **0.67**

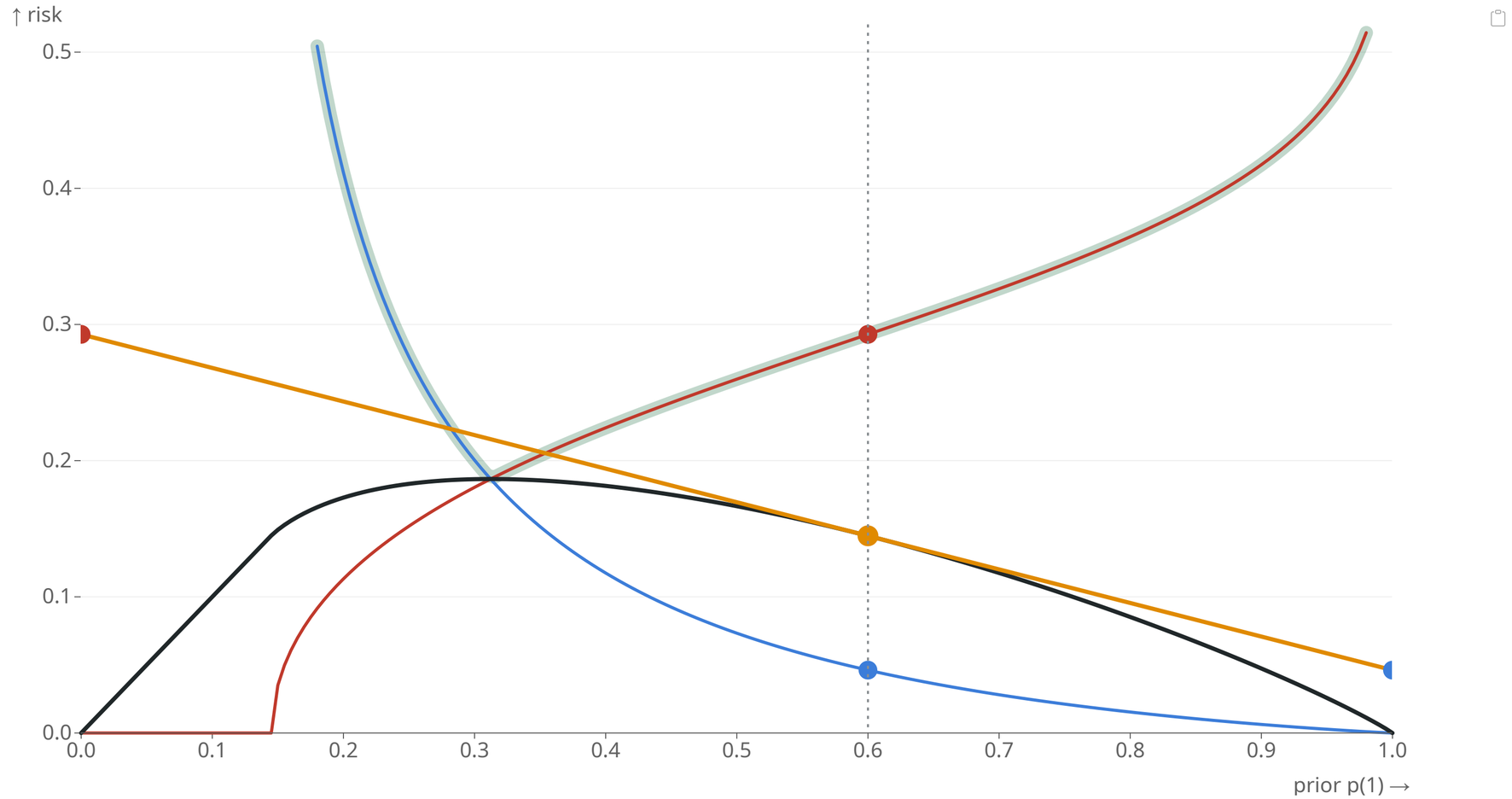
t_1 = **-10.70**

t_2 = **-1.59**

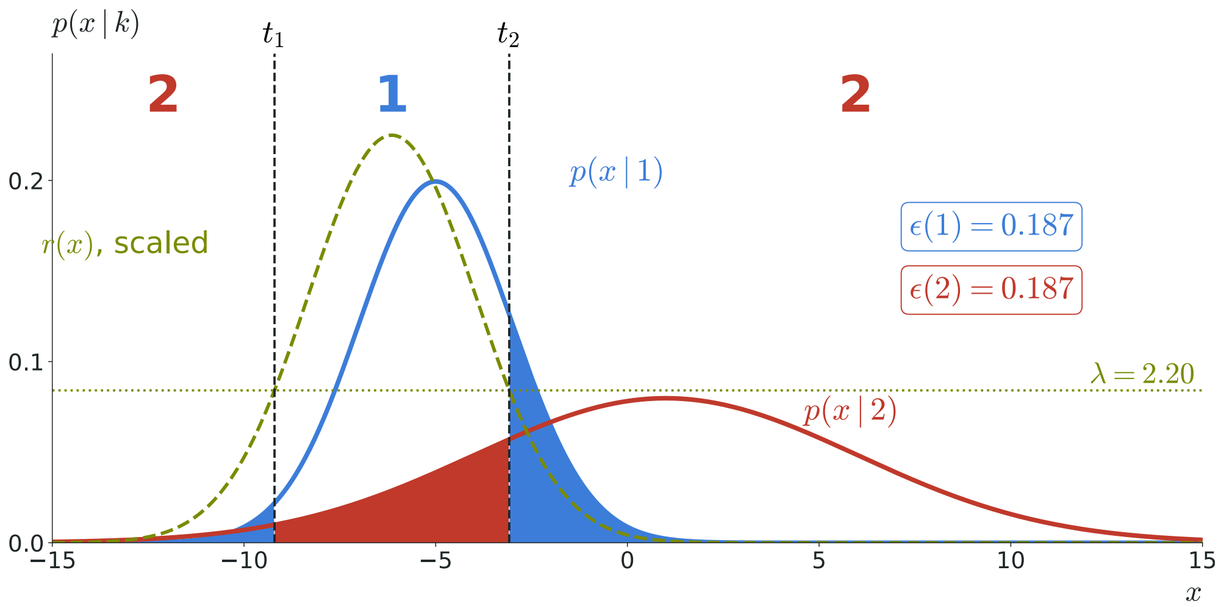
$\varepsilon(1)$ = **0.046**

$\varepsilon(2)$ = **0.293**

max = **0.293**



The minimax strategy for example III



Search over the prior instead of λ

```
1 from scipy.optimize import fminbound
2 p1 = fminbound(worst_risk, 0, 1)
```

$$p^*(1) = 0.312, \quad \lambda^* = 2.20$$

$$t_1 = -9.20, \quad t_2 = -3.08$$

$$\epsilon(1) = \epsilon(2) = \mathbf{0.187}$$

$$q^*(x) = \begin{cases} 1 & \text{if } t_1 < x < t_2 \\ 2 & \text{otherwise} \end{cases}$$

A grid of priors is **too coarse**.

Output of the assignment

Part 1: discrete

- `minimax_strategy_discrete`: strategy, i^* and the errors for all $N + 1$ thresholds
- `classify_discrete`: labels for test images

Part 2: continuous

- `risk_fix_q_cont`: fixed strategy, all priors
- `worst_risk_cont`: worst risk for one prior
- `minimax_strategy_cont`: the result



Copy your Lab 2 functions into `minimax.py`. Imports do not work in the BRUTE evaluation.

Take-away for the lab

- ☑ **Minimax** needs only $p(x \mid k)$: no priors, no loss
- ☑ The strategy is a **threshold on the likelihood ratio** $r(x)$
- ☑ **Discrete x:** sort by r, try all $N+1$ thresholds, take the smallest maximum
- ☑ **Continuous x:** the optimum is where $\epsilon(1) = \epsilon(2)$
- ☑ **Normal distributions:** minimax is the Bayesian strategy for the **worst prior**
- ☑ **Next week: test!** Bring a calculator


"That's all Folks!"